



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院



Chapter 1

INTRODUCTION

Course Catalog



1. Introduction
2. The Physical Layer
3. The Data Link Layer
4. The Medium Access Control Sub-Layer
5. The Network Layer
6. The Transport Layer
7. The Application Layer
8. Network Security

DLL: Data Link Layer

- LLC: Logical Link Control
- MAC: Medium Access Control

Chapter 7 Application

Chapter 6 Transport

Chapter 5 Network

Chapter 3 LLC

Chapter 4 MAC

Data Link

Chapter 2 Physical

Contents



1. Uses of Computer Networks
2. Network Hardware
3. Network Software
4. Reference Models
5. Example Networks
6. Network Standardization
7. Metric Units
8. Outline of the Rest of the Book

Basic Concepts



➤ Computer Network

- ✦ a collection of **autonomous** computers **interconnected** by a single technology.

➤ Internet

- ✦ **Inter**connected **Networks**.

➤ the Internet

- ✦ **the** most well-known **network** of internetworks.

➤ Distributed System

- ✦ a collection of independent computers **appears** to its users as a single coherent system.

autonomous: 自主的; 自治的

coherent: 一致的, 连贯的

1.1 Uses of Computer Networks



1. Business Applications
2. Home Applications
3. Mobile Users
4. Social Issues

1.1.1 Business Applications



- Resource Sharing
 - ✦ printer, information, ...
- Remote Accessing
 - ✦ VPN: Virtual Private Networks
- Client-Server model
 - ✦ Data Sharing
 - ✦ VoIP: Voice over IP
 - ✦ IP Telephone
 - ✦ Desktop sharing
 - ✦ E-Commerce: electronic commerce

Client and Server (C/S) Model

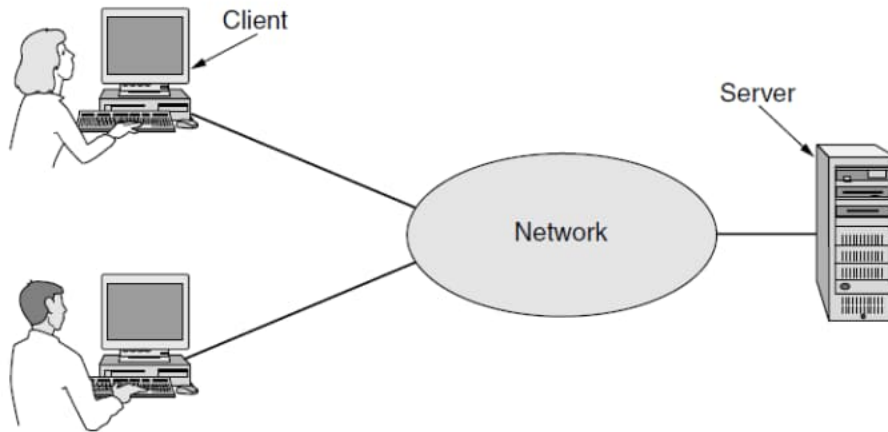


Figure 1-1. A network with two clients and one server.

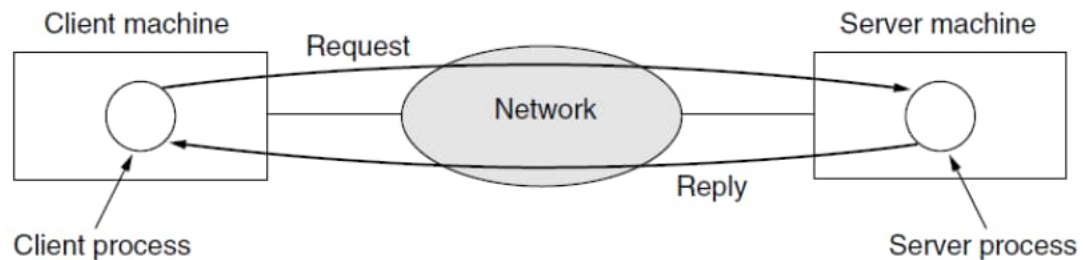


Figure 1-2. The client-server model involves requests and replies.

1.1.2 Home Applications

- Access to remote information
 - ✦ Reading News, Online digital library, Instant messages, Telelearning (Youtube, 小红书, Bilibili, ...)
- Instant Message
 - ✦ WhatsApp, Facebook Messenger, WeChat, Instagram, Snapchat, Viber, Line, Telegram, ...
- Social Network
 - ✦ Facebook, X (Twitter, tweet), Wikipedia, ..., 抖音, 微博, 朋友圈, ...
- Electronic Commerce
- Entertainment
 - ✦ Music, IP TeleVision, Game Playing, ...
- Ubiquitous Computing
 - ✦ Sensors, power-line networks, RFIDs, ...

tweet: 小鸟的吱喳声

Ubiquitous: 无处不在的, 普适的

Peer-to-Peer system

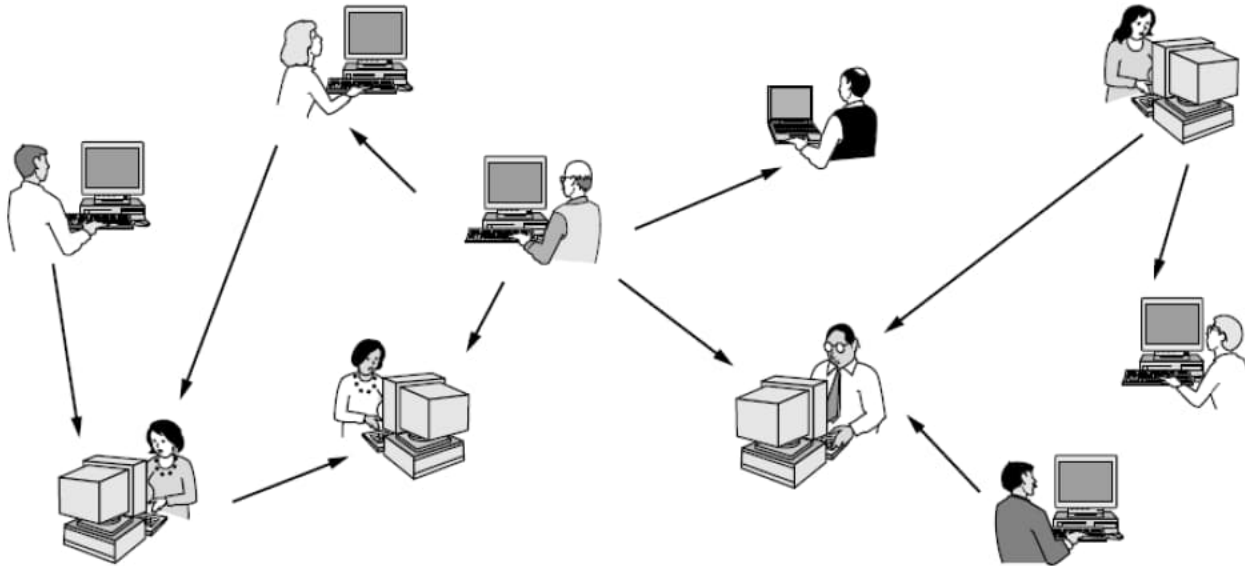


Figure 1-3. In a peer-to-peer system there are **no fixed** clients and servers.

Some forms of e-commerce



Tag	Full name	Example
B2C	Business-to-consumer	Ordering books online
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products online
P2P	Peer-to-peer	Music sharing
P2P	Person to person	Financial Deposit and loan (China)

Figure 1-4. Some forms of e-commerce.

1.1.3 Mobile Users

- Wireless Networks
- Navigator
 - ✦ GPS(Global Positioning System) + Smart phones
- Mobile-Commerce
 - ✦ e-Payment (NFC: Near Field Communication)
- Sensor Networks
 - ✦ Cars
- Wearable Computers:
 - ✦ Smart Watch
 - ✦ Implantable Devices: Pacemaker, Insulin pumps, ...

pacemaker: 心脏起搏器

insulin: 胰岛素

Wireless and Mobile



Wireless	Mobile	Typical applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in unwired buildings
Yes	Yes	Store inventory with a handheld computer

Figure 1-5. Combinations of wireless networks and mobile computing.

inventory : 清查, 盘点

1.2 Network Hardware



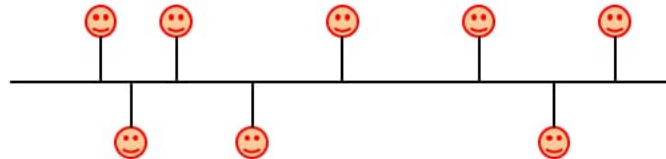
1. Personal Area Networks
2. Local Area Networks
3. Metropolitan Area Networks
4. Wide Area Networks
5. Internetworks
 - The Internet

Two types of Transmission Technology



➤ Broadcast links

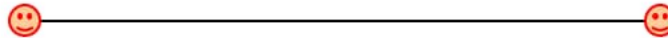
- ✦ Communication channel shared by **all machines**
- ✦ **Packets** sent by any machine are received by all the others.
- ✦ Broadcast systems usually also allow the possibility of **addressing** a packet to **all** destinations by using a special code in the address field.



packet: short message, 分组
message: 报文
addressing: 编址, 寻址

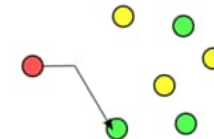
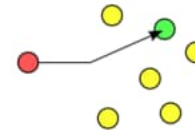
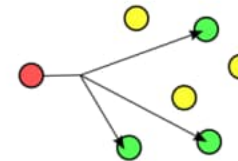
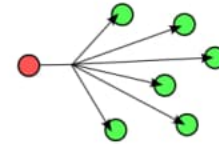
➤ Point-to-Point links

- ✦ connect **individual pairs** of machines.
- ✦ To go from the source to the destination, packets may have to visit one or more **intermediates** machines.
- ✦ Often multiple routes, of different lengths, are possible, so **finding good ones** is important in point-to-point networks.



Packet Transferring Mode

- **Broadcast** (广播): one-to-**all**, transferring a message to all recipients simultaneously.
- **Multicast** (多播/组播): one-to-**group**, data transmission is addressed to a group of destination computers simultaneously.
- **Unicast** (单播): one-to-**one**, from one point to another point.
- **Anycast** (任播/泛播): one-to-**one-of-group**, a single destination address has multiple routing paths to two or more endpoint destinations.
- **Geocast** (区域多播): one-to-**group**, by their **geo**graphical locations.



Classification by scale



Interprocessor distance	Processors located in same	Example	
1 m	Square meter	Personal area network	PAN
10 m	Room	Local area network	LAN
100 m	Building		
1 km	Campus		
10 km	City	Metropolitan area network	MAN
100 km	Country	Wide area network	WAN
1000 km	Continent		
10,000 km	Planet	The Internet	The Internet
> 100,000 km	Solar System	Inter Planet Network	IPN

Figure 1-6. Classification of interconnected processors by scale.

1.2.1 Personal Area Networks

- **PANs** (Personal Area Networks)
 - ✦ let devices communicate over the range of a person.

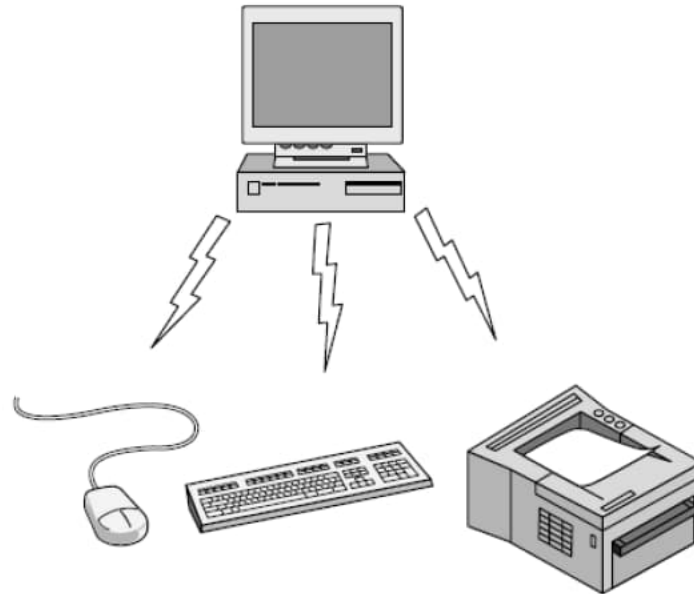


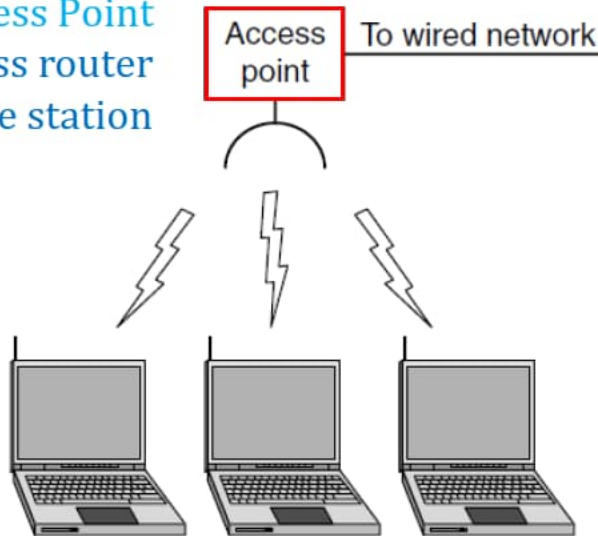
Figure 1-7. Bluetooth PAN configuration.

1.2.2 Local Area Networks

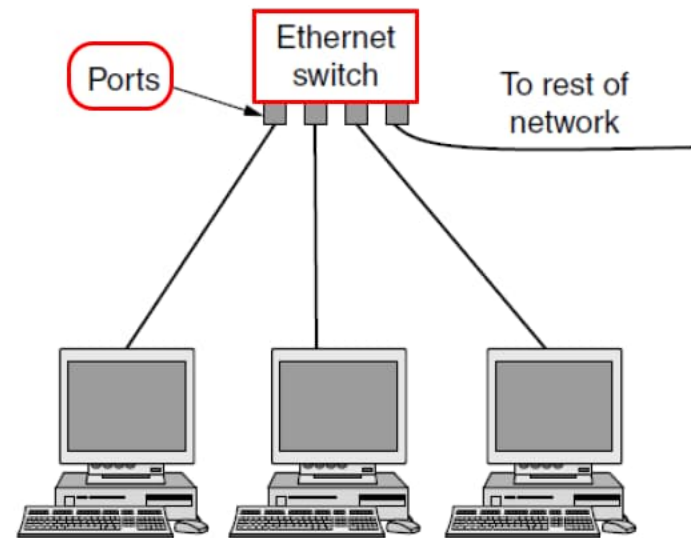
➤ A LAN

- ★ is a privately owned network that operates within and nearby a single building like a home, office or factory.

AP: Access Point
Wireless router
Base station



(a) IEEE 802.11



(b) Switched Ethernet.

Figure 1-8. Wireless and wired LANs.

1.2.3 Metropolitan Area Networks

- A **MAN** (Metropolitan Area Network) covers a city.

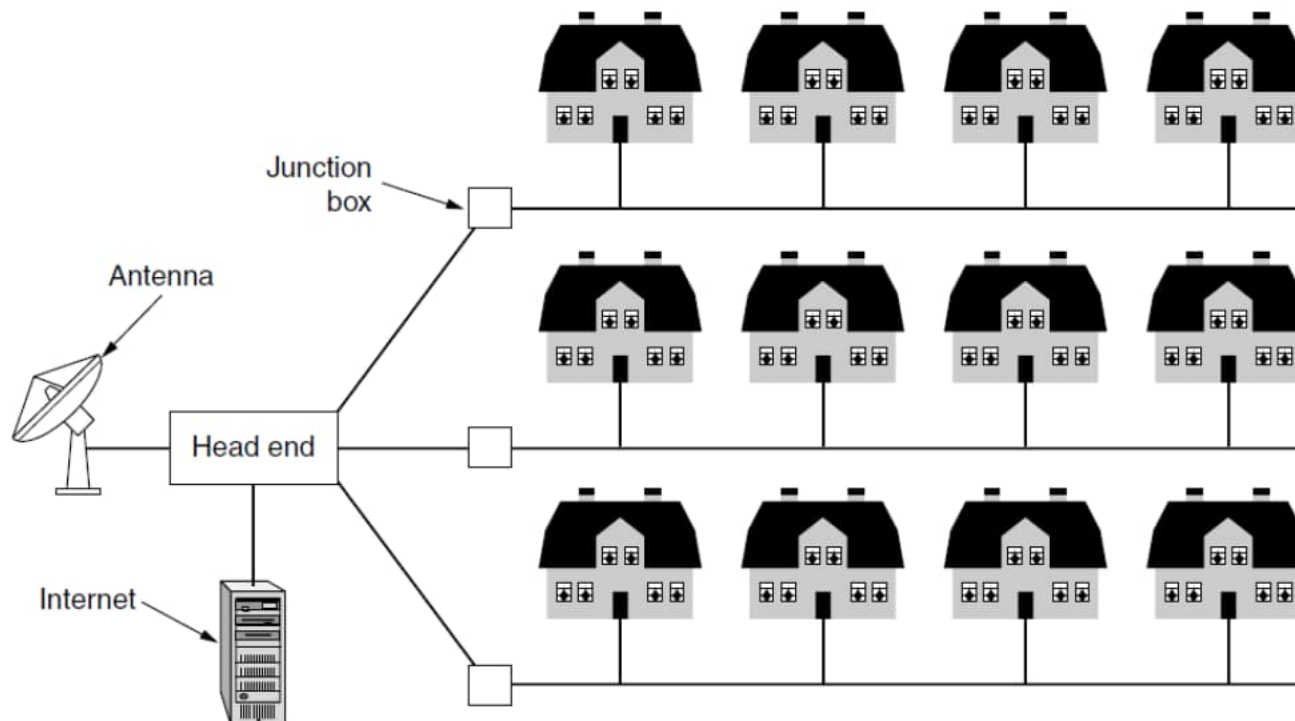


Figure 1-9. A metropolitan area network based on cable TV.

1.2.4 Wide Area Networks



➤ A WAN (Wide Area Network)

- ✦ spans a large geographical area, often a country or continent.

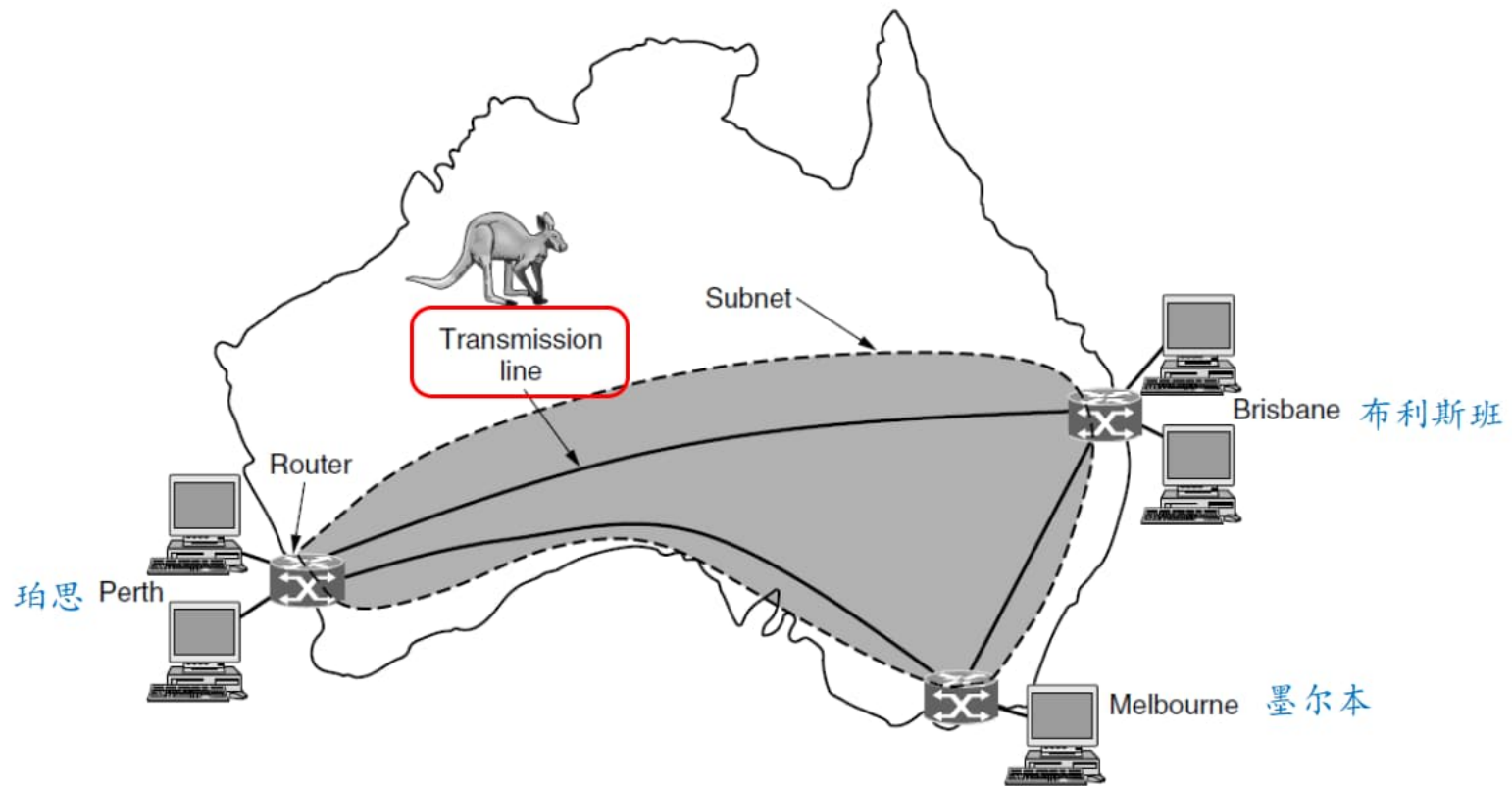


Figure 1-10. WAN that connects three branch offices in Australia.

WAN using a VPN



VPN: Virtual Private Network

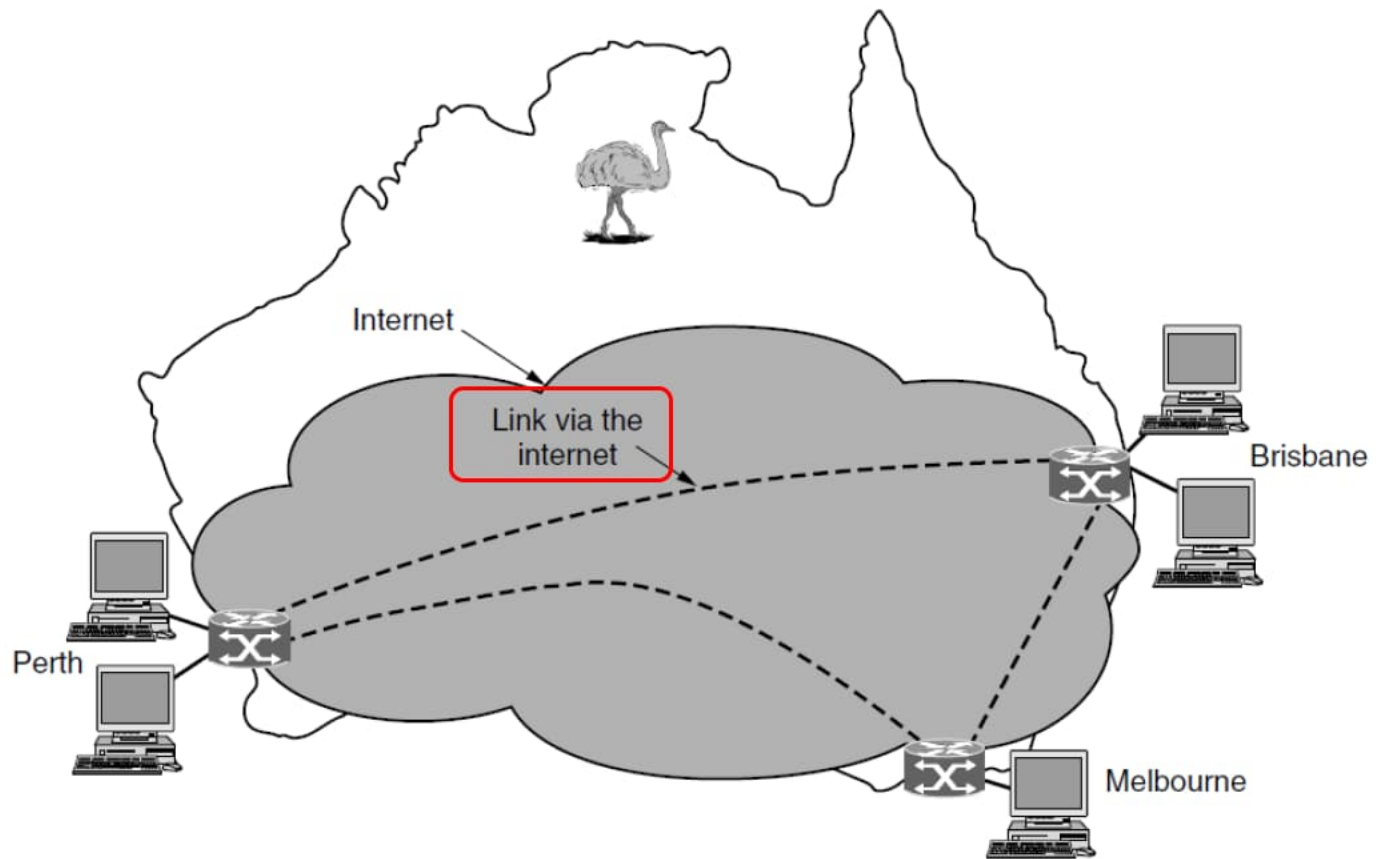


Figure 1-11. WAN using a virtual private network.

WAN using an ISP network



ISP: Internet Service Provider

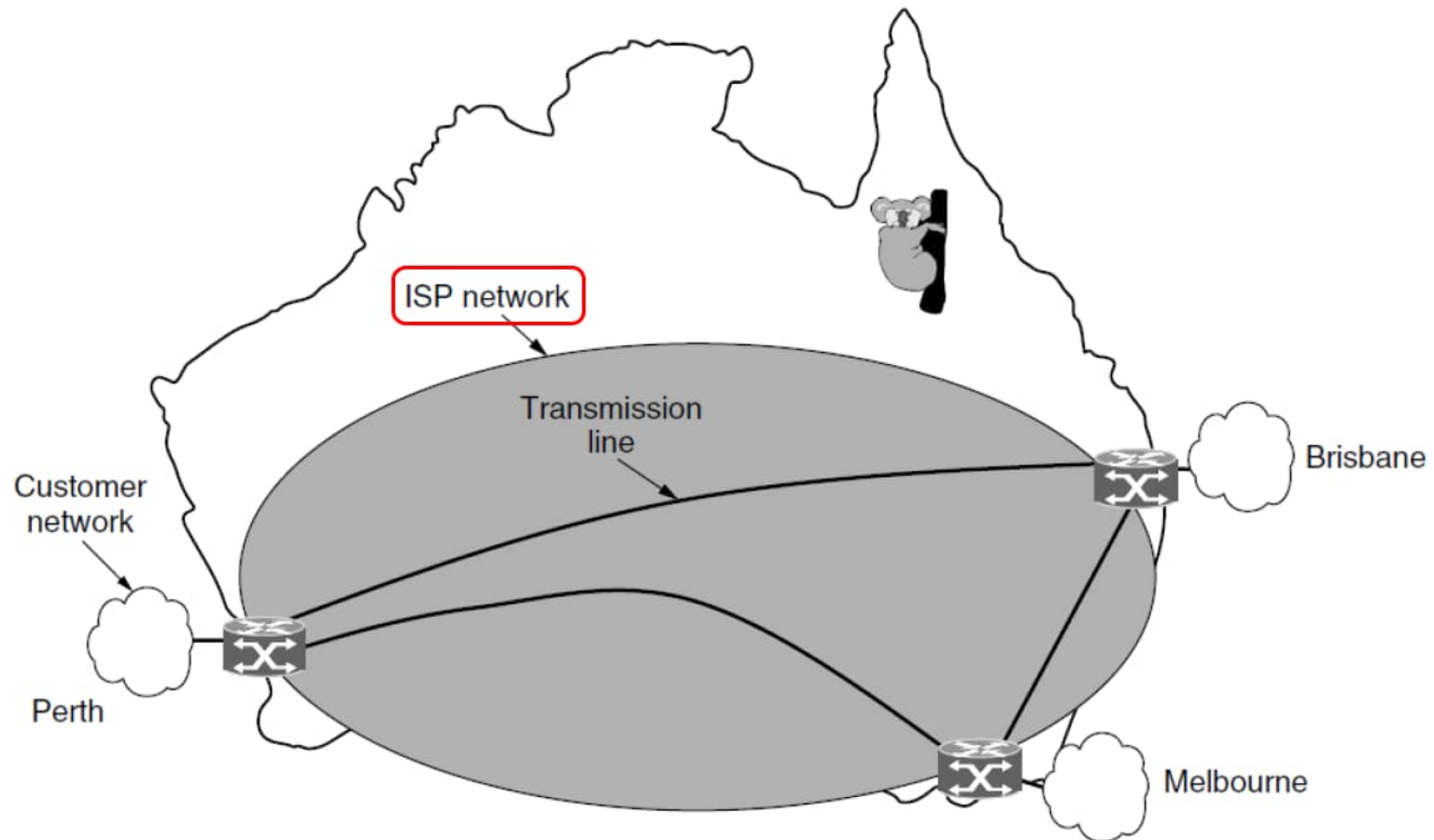


Figure 1-12. WAN using an ISP network.

1.2.5 Internetworks

➤ Difference between **internetwork** and **the Internet**

- ✦ **Internetwork:** A collection of interconnected networks.
 - It is called an **internetwork** or **internetwork**.
- ✦ **The Internet:** the most widespread internetwork over the globe we are surfing.

internetwork, internetwork, 网际, 网间网, 互联网
the Internet, 因特网

1.3 Network Software



1. Protocol Hierarchies
2. Design Issues for the Layers
3. Connection-Oriented Versus Connectionless Service
4. Service Primitives
5. The Relationship of Services to Protocols

1.3.1 Protocol Hierarchies

- Peers
 - ✦ the entities comprising the corresponding layers on different machines.
- Protocol
 - ✦ an **agreement** between the communicating parties on how communication is to proceed.
- Interface
 - ✦ defines which **primitive operations** and
 - ✦ **services** the lower layer makes available to the upper one.
- Protocol Stack (协议栈)
 - ✦ a **list of the protocols** used by a certain system, one protocol per layer
- Network Architecture
 - ✦ a **set of layers** and **protocols**

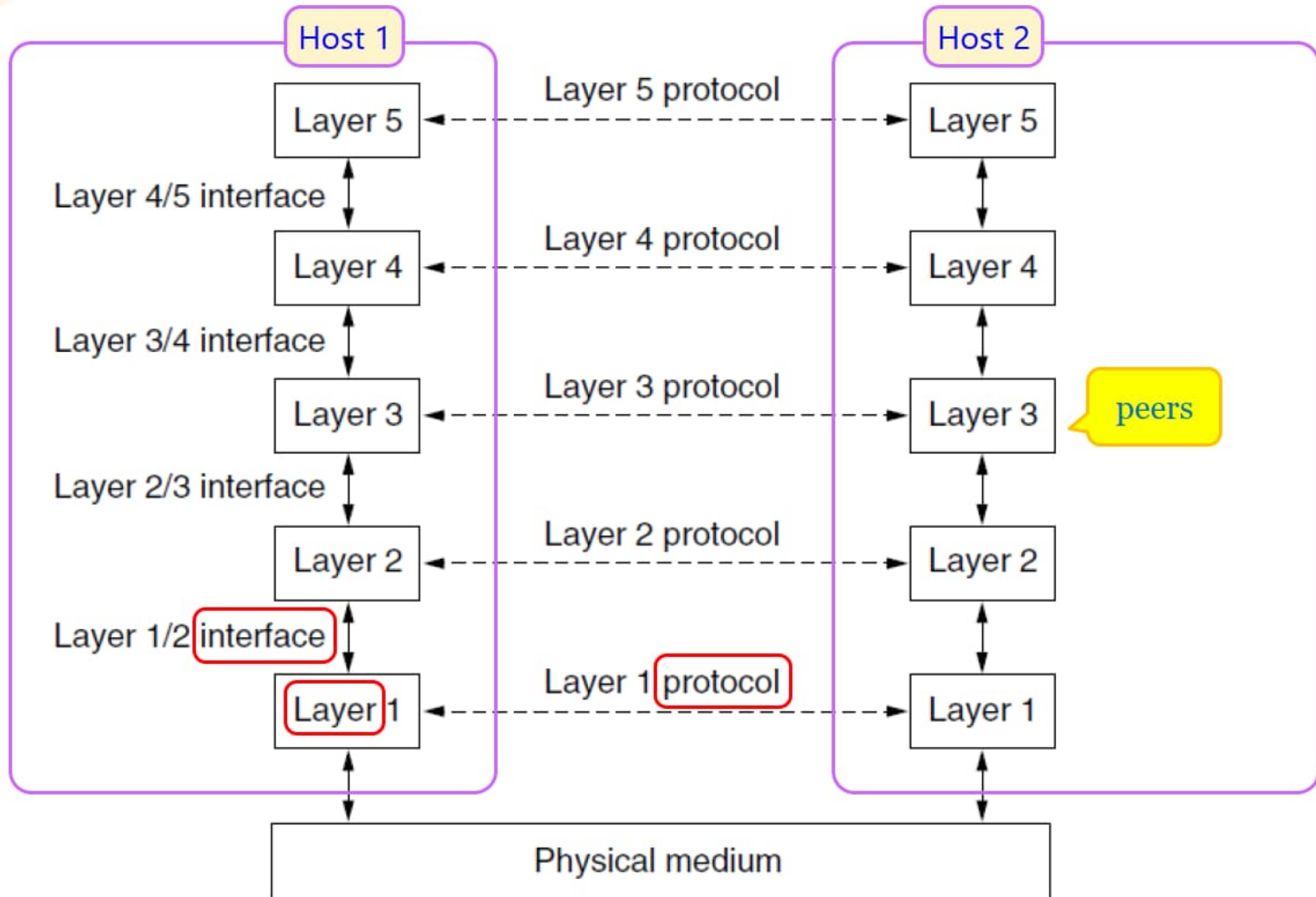


Figure 1-13. Layers, protocols, and interfaces.

An Example of Protocol Hierarchies

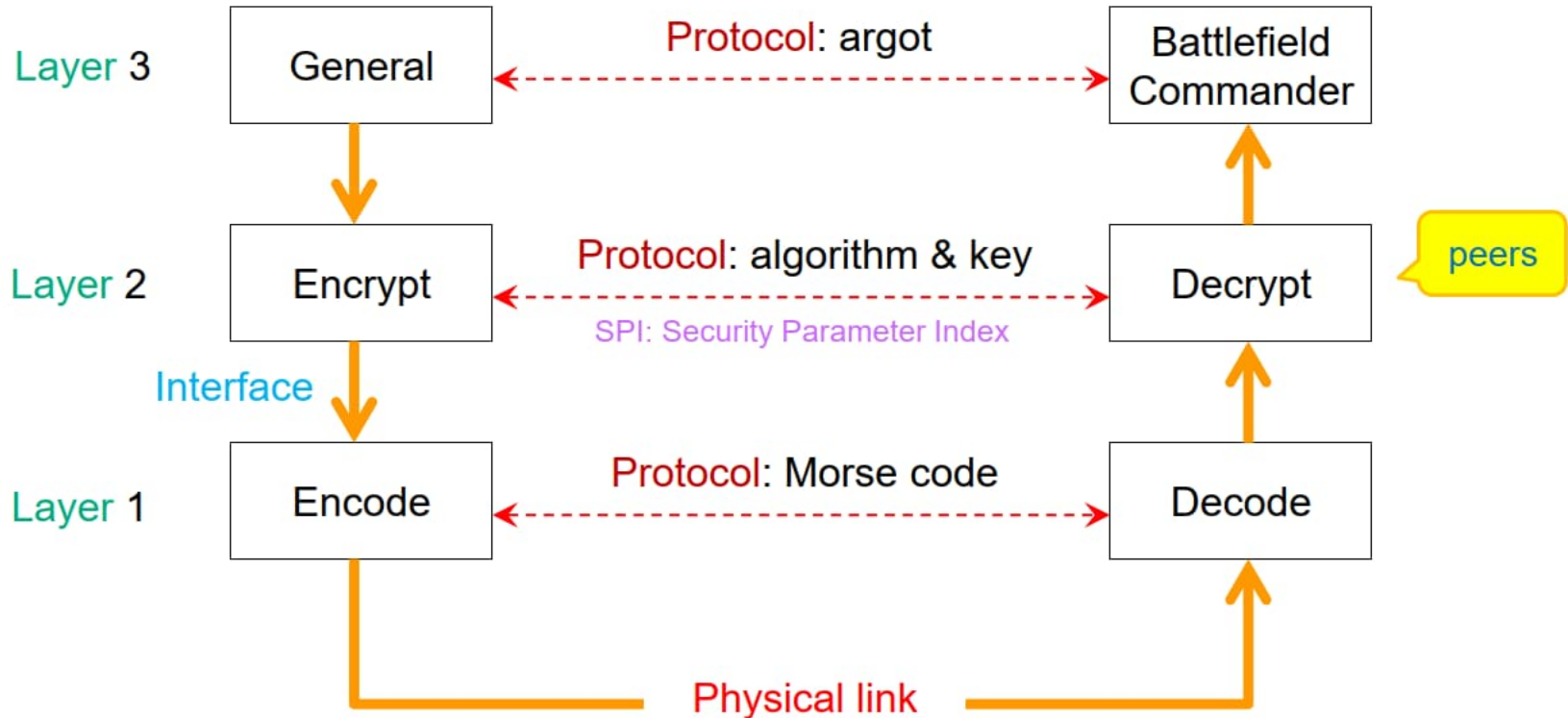


Figure 1-extra. Battlefield Communication

argot 英 ['ɑ:gəʊ] 美 ['ɑ:rgət] n. 行话, 黑话, 暗语

Virtual Communication

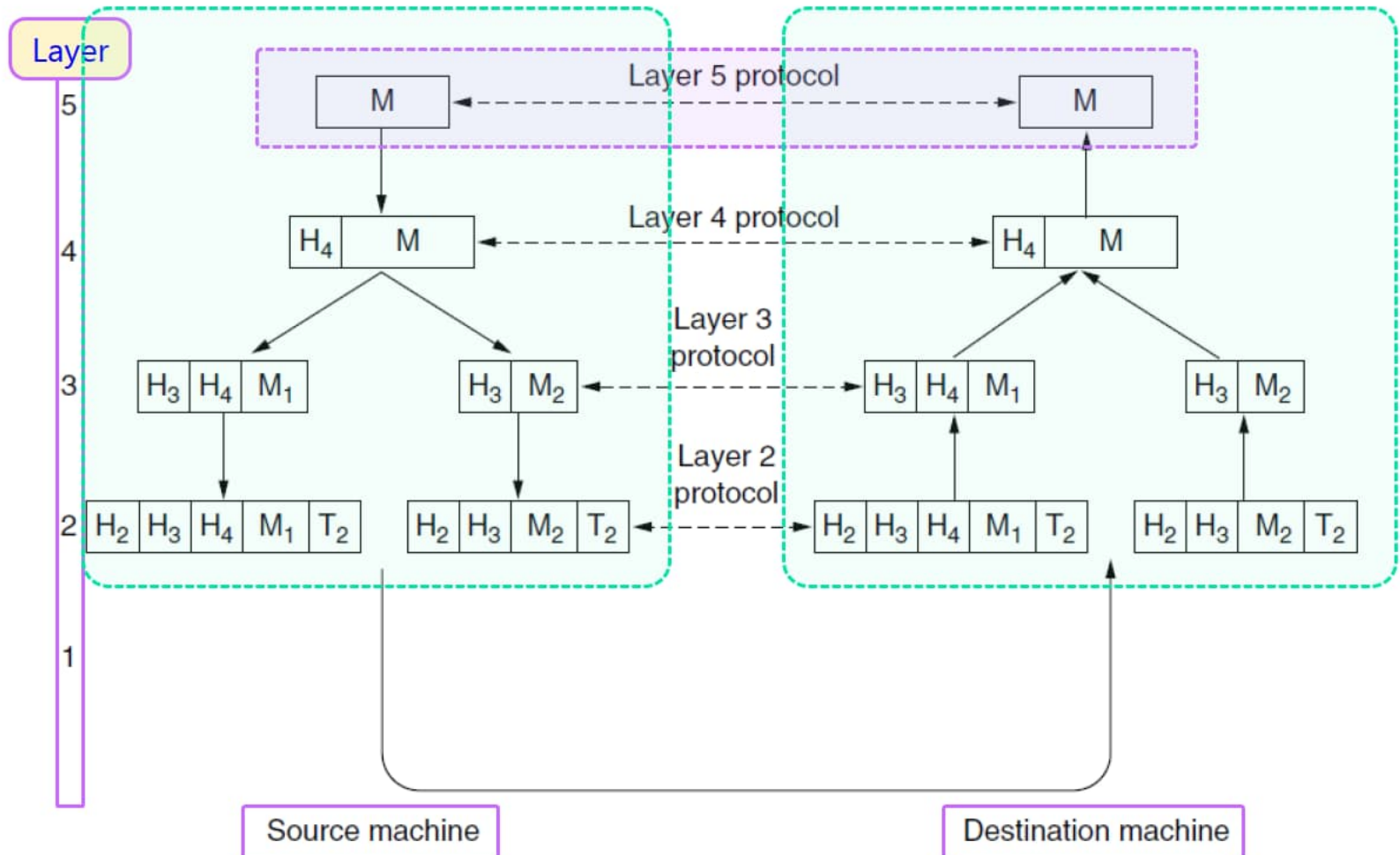


Figure 1-15. Example information flow supporting virtual communication in layer 5.

1.3.2 Design Issues for the Layers

- Reliability
 - ✦ making a network that **operates correctly**.
- Protocol Layering
 - ✦ **support change** by dividing the overall problem and
 - ✦ **hiding implementation details**.
- Resource Allocation
 - ✦ divide their resources so that one host **does not interfere with another** too much.
- Security
 - ✦ **defending** against different kinds of **threats**.

Reliability: 可靠性

Protocol Layering: 协议分层

Resource Allocation: 资源分配

Security: 安全

Reliability



- Error Control: **find** and **fix** errors.
 - ✦ **Error detection**: Information that is incorrectly received can be retransmitted until it is received correctly.
 - ✦ **Error correction**: the **correct** message is recovered from the possibly **incorrect** bits that were originally received.
- Routing: finding a working **path** through a network.
 - ✦ Static
 - ✦ Dynamic

Error Control: 差错控制

Error detection: 检错

Error correction: 纠错

Routing: 路由选择

Protocol Layering



- Addressing or Naming

- ✦ a mechanism for **identifying** the senders and receivers.

- Internetworking

- ✦ different network technologies often have different limitations.

- Scalable

- ✦ enable network get large

Addressing: 编址, 寻址

Naming: 命名

Internetworking: 网络互联

Scalable: 可扩展

Resource Allocation

- Statistical multiplexing
 - ✦ sharing resource based on the statistics of demand
- Flow Control
 - ✦ to keep a fast sender **from swamping** a slow receiver with data
- Congestion Control
 - ✦ overloading of networks
- Quality of Service
 - ✦ description or measurement of the overall performance of a service
 - packet loss, bit rate, throughput, transmission delay, availability, jitter, ...

Statistical multiplexing: 统计复用

Flow Control: 流量控制

Congestion Control: 拥塞控制

Quality of Service: 服务质量

➤ Confidentiality

- ✦ defend against eavesdropping on communications.

➤ Authentication

- ✦ prevent someone from impersonating someone else.

➤ Integrity

- ✦ prevent surreptitious changes to messages.

➤ Nonrepudiation

- ✦ no party to a contract can later deny having signed it

Confidentiality: 保密

Authentication: 认证

Integrity: 完整

Nonrepudiation: 不可抵赖

eavesdrop 英 ['i:vzdrɒp] 美 ['i:vzdra:p] vi. 偷听 (别人的谈话)

impersonate 英 [ɪm'pɜ:sənert] 美 [ɪm'pɜ:rsənert] vt. 模仿; 扮演;

surreptitious 英 [ˌsʌrəp'tɪʃəs] 美 [ˌsɜ:rəp'tɪʃəs] adj. 鬼鬼祟祟的; 私下的, 秘密的

1.3.3 Connection-Oriented Versus Connectionless Service

➤ Connection-oriented service

- ✦ establishes a connection.
- ✦ uses the connection.
- ✦ releases the connection.

➤ Connectionless service

- ✦ each message carries the full destination address.
- ✦ each one is routed through the intermediate nodes.

Connection-oriented service: 面向连接的服务

Connectionless service: 无连接服务

Packet switching modes



➤ Packet

- ✦ a short message at the network layer.

➤ store-and-forward switching

- ✦ receive a message in full before sending it

➤ cut-through switching

- ✦ transmission of a message before it is completely received

store-and-forward switching: 存储转发

cut-through switching: 直通转发

different types of service



	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

Figure 1-16. Six different types of service

1.3.4 Service Primitives



➤ Primitives (Operations)

- ★ If the protocol stack is located in the operating system, as it often is, the primitives are normally **system calls**.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Figure 1-17. Six service primitives that provide a simple connection-oriented service.

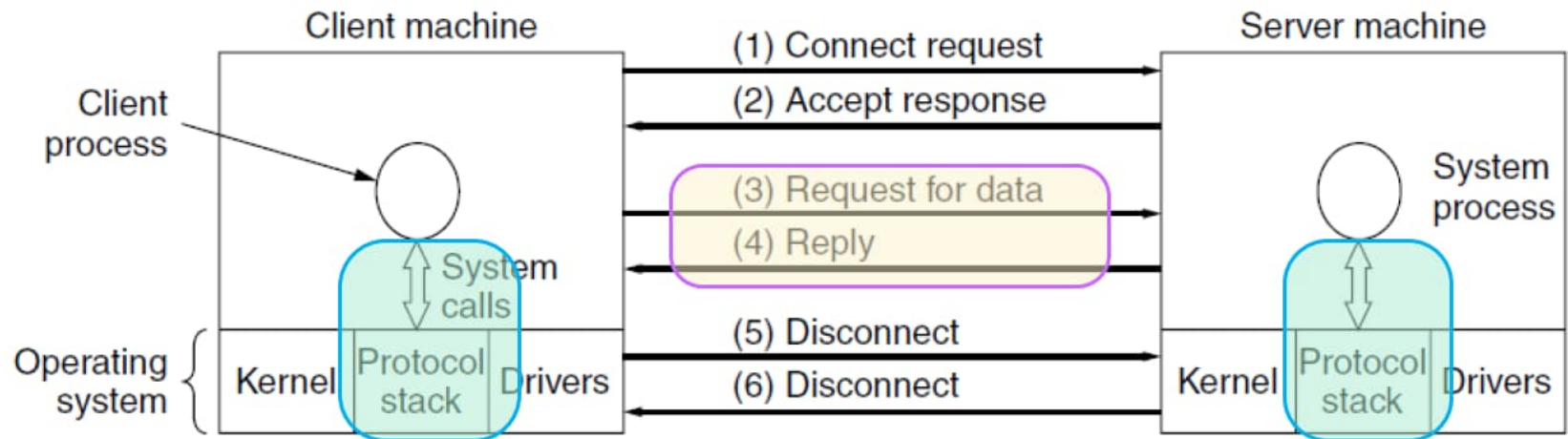


Figure 1-18. A simple client-server interaction using acknowledged datagrams.

1.3.5 The Relationship of Services to Protocols

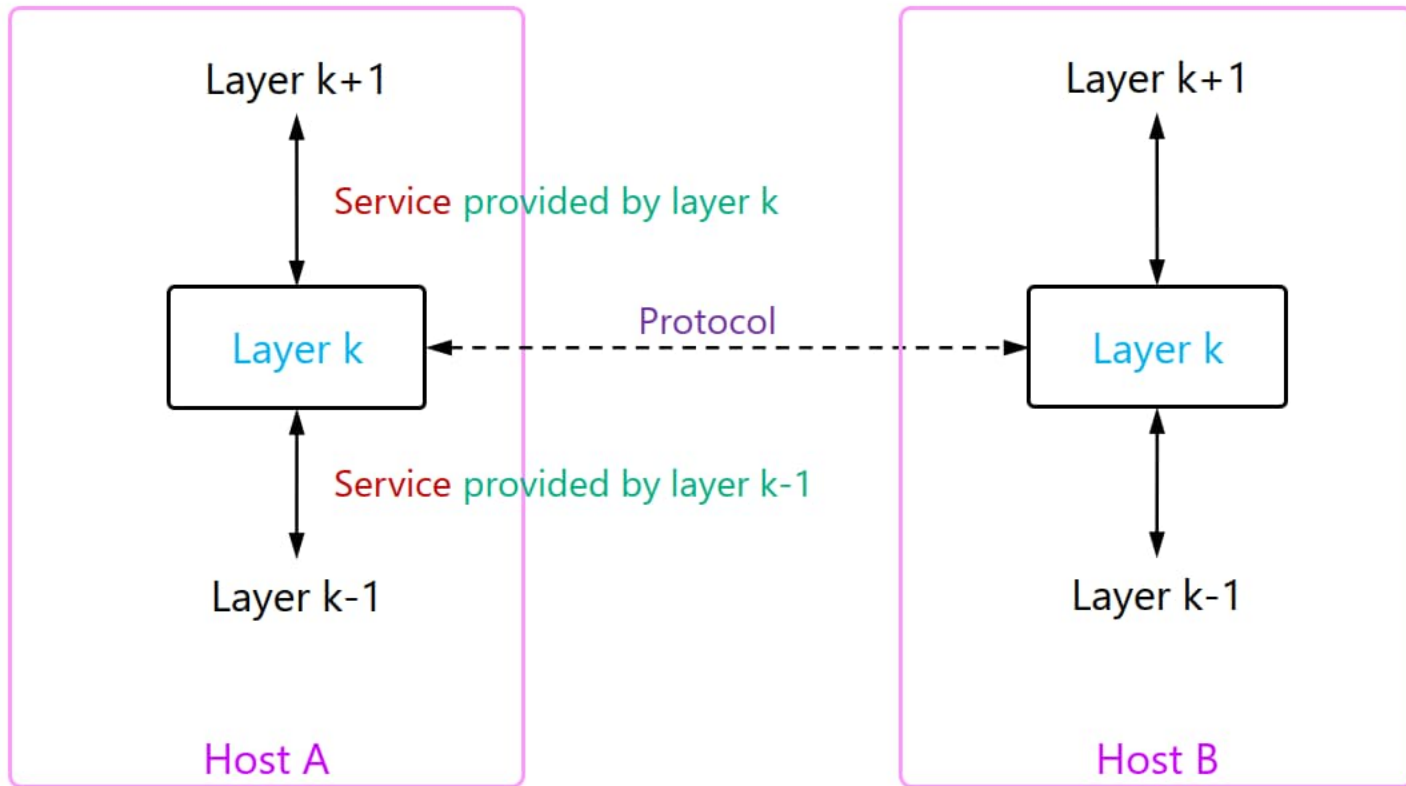


Figure 1-19. The relationship between a **service** and a **protocol**.

1.4 Reference Models



1. The OSI Reference Model
2. The TCP/IP Reference Model
3. The Model Used in This Book
4. A Comparison of the OSI and TCP/IP Reference Models
5. A Critique of the OSI Model and Protocols
6. A Critique of the TCP/IP Reference Model

1.4.1 The OSI Reference Model

- ISO/OSI RM (the **OSI model** for short)
 - ✦ **ISO**: International Standards Organization
 - ✦ **OSI**: Open Systems Interconnection
 - ✦ **RM**: Reference Model

➤ Principles

1. A layer should be created where a **different abstraction** is needed.
2. Each layer should perform a **well-defined** function.
3. The function of each layer should be chosen **with an eye toward** defining internationally standardized protocols.
4. The layer boundaries should be chosen to **minimize the information flow** across the interfaces.
5. The **number of layers** should be
 - a) **large enough** that distinct functions need not be thrown together in the same layer out of necessity, and
 - b) **small enough** that the architecture does not become unwieldy.

well-defined: 明确的(无歧义、无耦合)

with an eye toward: 着眼于

unwieldy: 笨拙的, 难以控制的

ISO/OSI RM

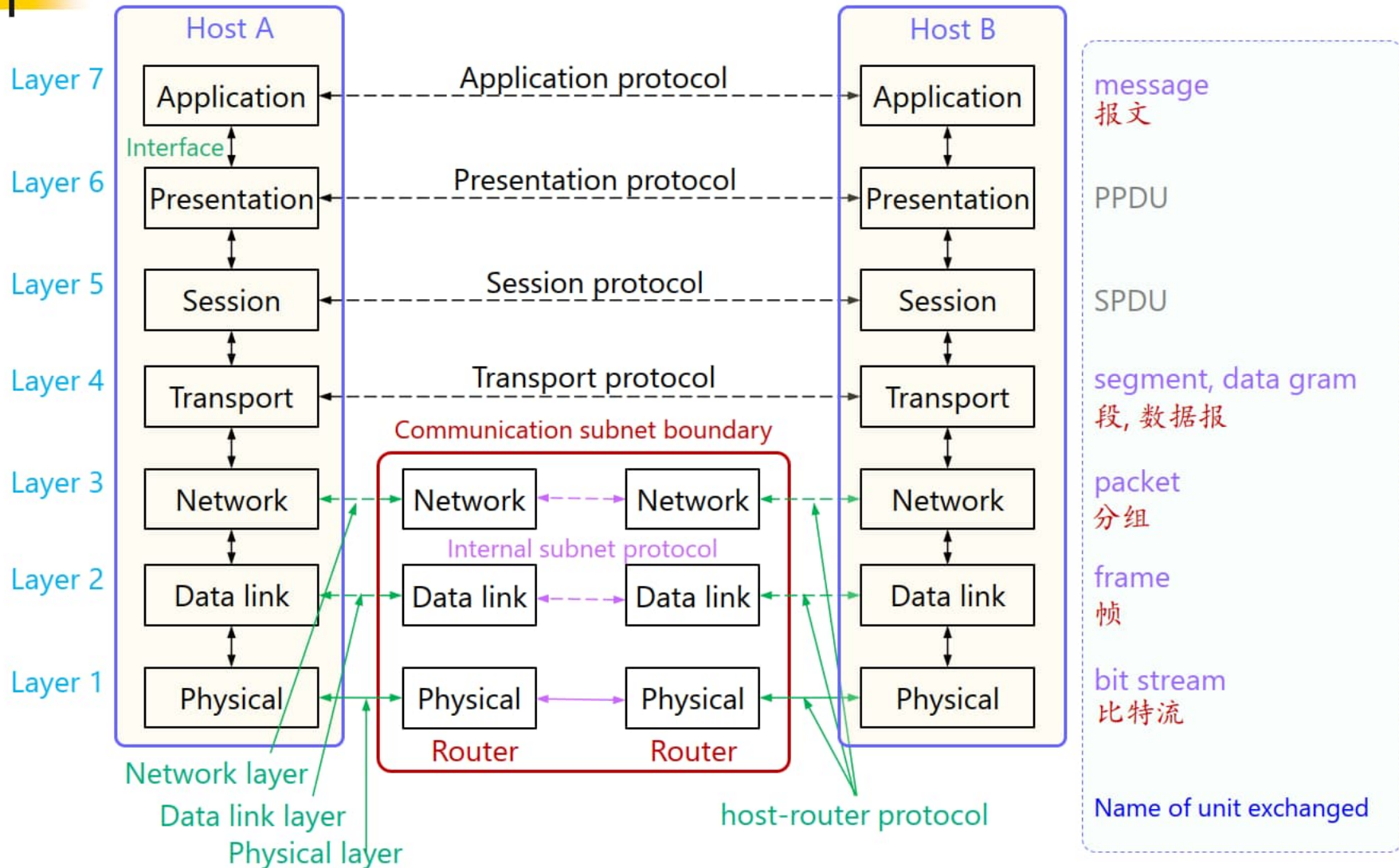


Figure 1-20. The OSI reference model.

1.4.2 The TCP/IP Reference Model

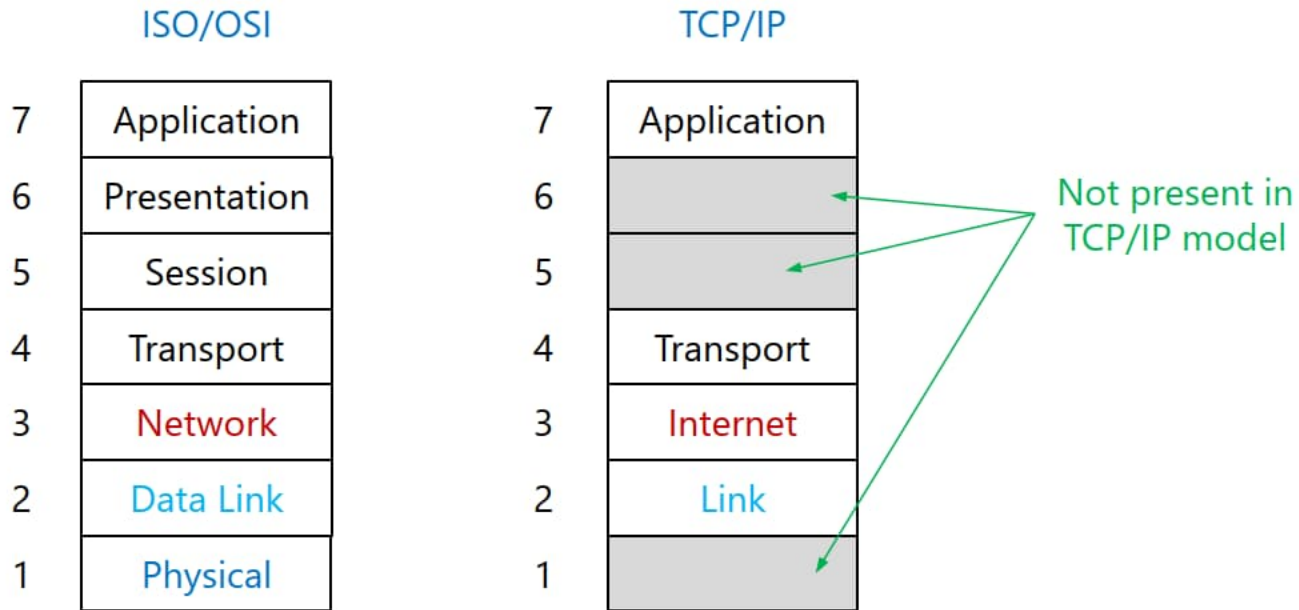


Figure 1-21. The TCP/IP reference model.

some important protocols

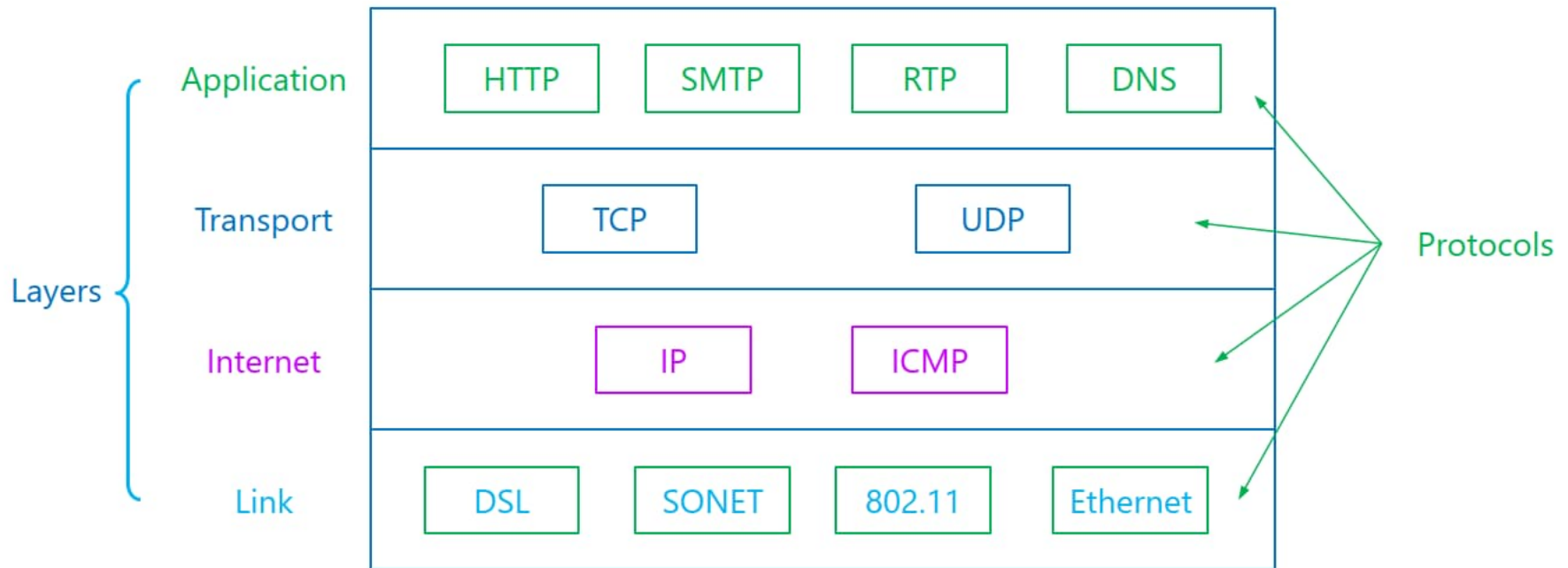


Figure 1-22. The TCP/IP model with some protocols we will study.

1.4.3 The Model Used in This Book

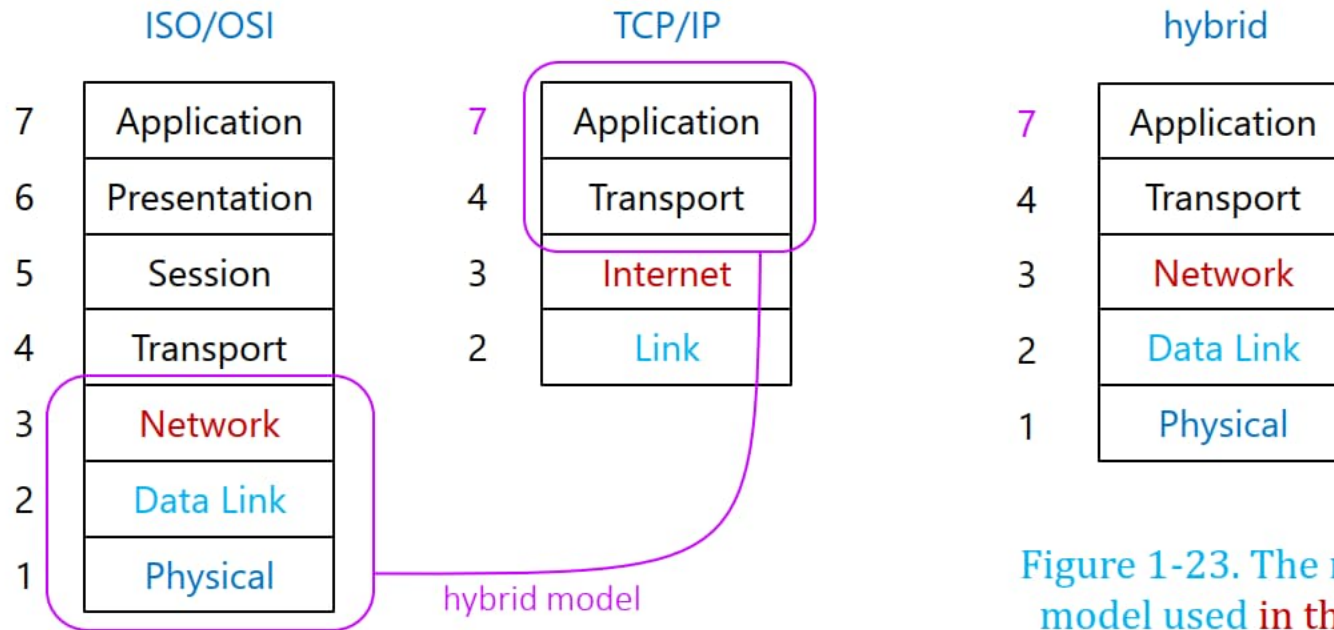


Figure 1-23. The reference model used in this book.

1.4.4 A Comparison of the OSI and TCP/IP Reference Models



➤ Similarities

- ✦ based on the concept of a stack of independent protocols
- ✦ the functionality of the layers is roughly similar
- ✦ the layers up through and including the transport layer are there to provide an end-to-end, network-independent transport service
- ✦ the layers above transport are application-oriented users of the transport service

Differences-1



➤ Services, Interfaces, & Protocols

- ★ Three concepts are **central** to the **OSI** model
 - Services: tells **what** the layer does. It defines the layer's **semantics**.
 - Interfaces: tells the processes above it **how** to access it.
 - Protocols: the **peer protocols** used in a layer are **the layer's own business**.
- ★ The **TCP/IP** model
 - **did not** originally clearly **distinguish** between **services, interfaces, and protocols**.

➤ Ordering

- ★ The OSI reference **model** was devised **before** the corresponding **protocols** were invented.
- ★ With TCP/IP the reverse was true: the **protocols** came **first**, and the **model** was really just a description of the existing protocols.

Differences-2



➤ Connectionless versus Connection-Oriented

★ The OSI model supports

- both **connectionless** and **connection-oriented** communication in the **network** layer,
- but **only connection-oriented** communication in the **transport** layer.

★ The TCP/IP model supports

- only one mode in the **network** layer (**connectionless**)
- but **both** in the **transport** layer, giving the users a choice.

	ISO/OSI	TCP/IP
Transport Layer	Connection-oriented	Both
Network Layer	Both	Connectionless

1.4.5 A Critique of the OSI Model and Protocols



- Bad timing.
- Bad technology.
- Bad implementations.
- Bad politics.

Bad Timing



➤ Apocalypse of the two elephants

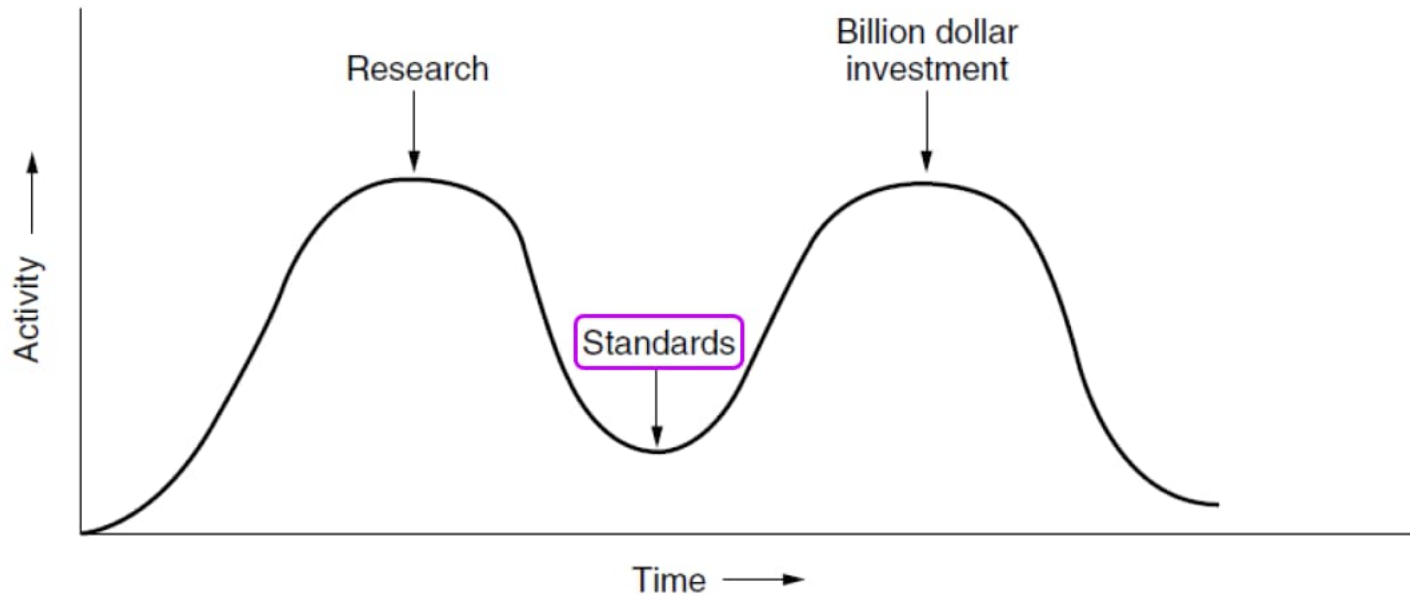


Figure 1-24. The apocalypse of the two elephants.

Apocalypse 英 [ə'pɒkəlɪps] 美 [ə'pɑkəlɪps]
n. 世界毁灭；大灾变；末世；末日启示

Bad Technology



- The choice of **seven** layers was **more political than technical**
 - ✦ two of the layers (session and presentation) are **nearly empty**
 - ✦ whereas two other ones (data link and network) are **overfull**
- extraordinarily **complex**
 - ✦ the printed standards occupy a significant fraction of a meter of paper
- **difficult** to implement
- **inefficient** in operation
- Some functions reappear **again and again** in each layer
 - ✦ such as: **addressing, flow control, and error control**

Bad Implementations



- the initial implementations were
 - ✦ huge, unwieldy, and slow.
- people to associate “OSI”
 - ✦ with “poor quality.”
- People began using TCP/IP quickly
 - ✦ which led to a large user community

Bad Politics



- TCP/IP was thought
 - ✦ as part of UNIX
 - and UNIX in the 1980's in academia was not unlike parenthood and apple pie.

- OSI, on the other hand, was widely thought to be
 - ✦ the **creature** of the European telecommunication ministries, the European Community,
 - ✦ and later the U.S. Government.

1.4.6 A Critique of the TCP/IP Reference Model

- The model does **not clearly** distinguish the concepts of
 - ✦ **services, interfaces, and protocols.**
- the TCP/IP model is **not** at all **general** and is **poorly suited** to describing any protocol stack other than TCP/IP.
 - ✦ Trying to use the TCP/IP model to describe Bluetooth, for example, is completely impossible.
- the **link layer** is not really a layer at all
 - ✦ It is an **interface** (between the network and data link layers).
- the TCP/IP model does not distinguish
 - ✦ between the **physical** and **data link** layers.
- The protocol implementations were distributed **free**,
 - ✦ widely used, deeply entrenched, and thus **hard to replace**.

entrenched 英 [ɪn'trentʃt] 美 [ɪn'trentʃt]
adj. 根深蒂固的, 牢固的, 固守的;

1.5 Example Networks



1. The Internet
2. Third-Generation Mobile Phone Networks
3. Wireless LANs: 802.11
4. RFID and Sensor Networks

1.5.1 The Internet



➤ The ARPANET

✦ Advanced Research Projects Agency

➤ IMP

✦ Interface Message Processor

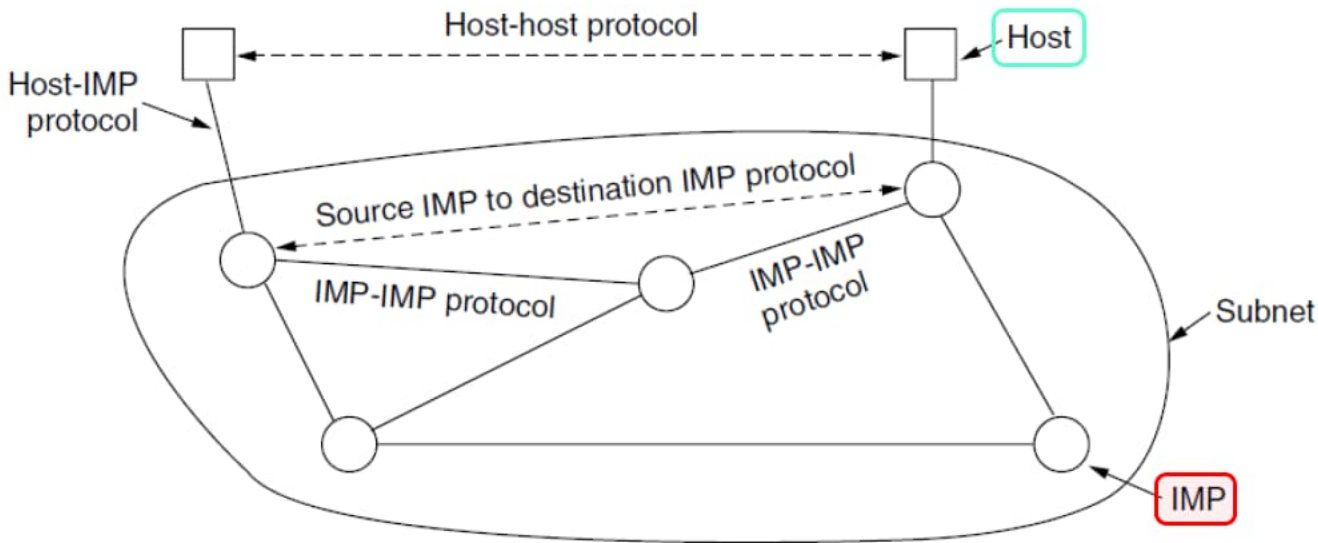
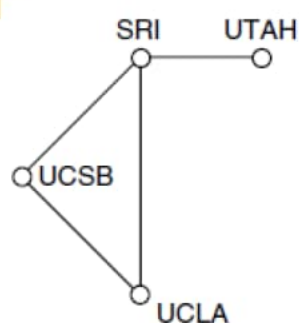


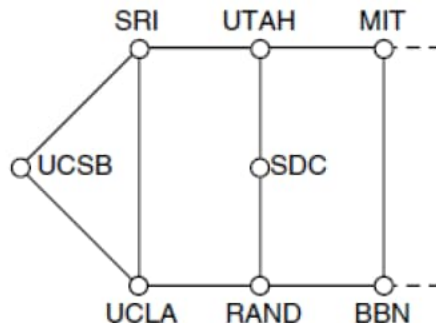
Figure 1-26. The original ARPANET design.

Growth of the ARPANET



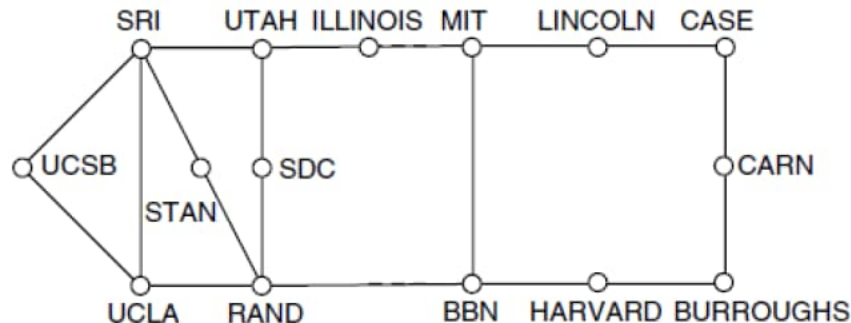
(a)

(a) December 1969.



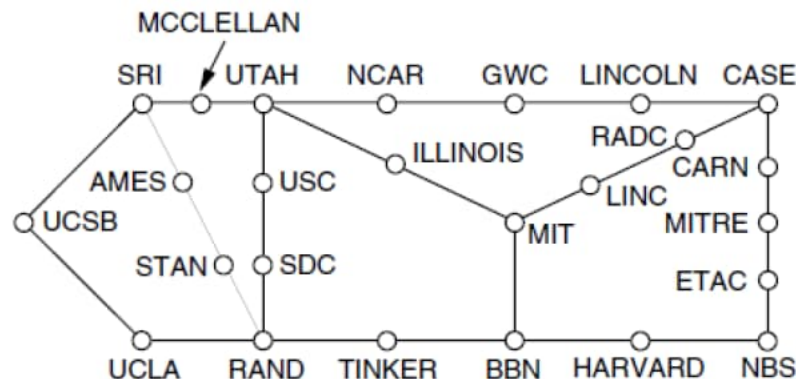
(b)

(b) July 1970.



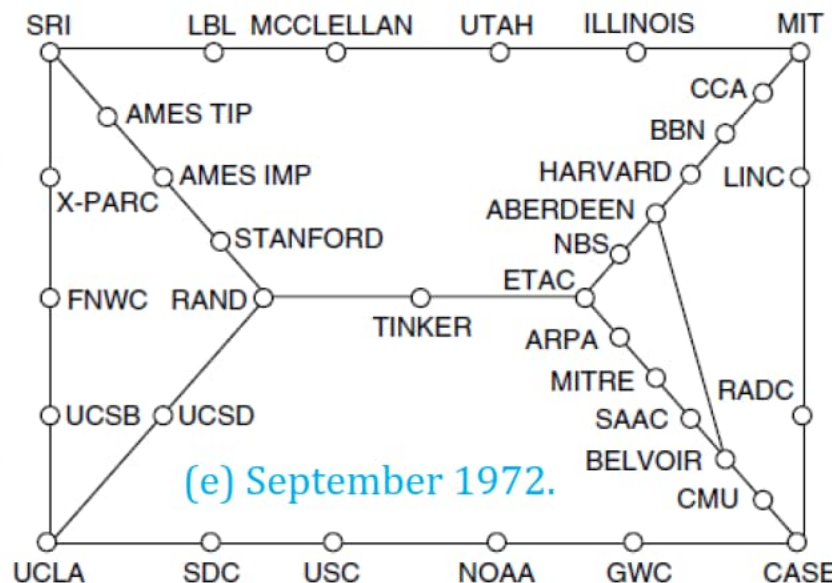
(c)

(c) March 1971



(d) April 1972.

(d)



(e) September 1972.

(e)

NSFNET



➤ NSF

✦ the U.S. National Science Foundation

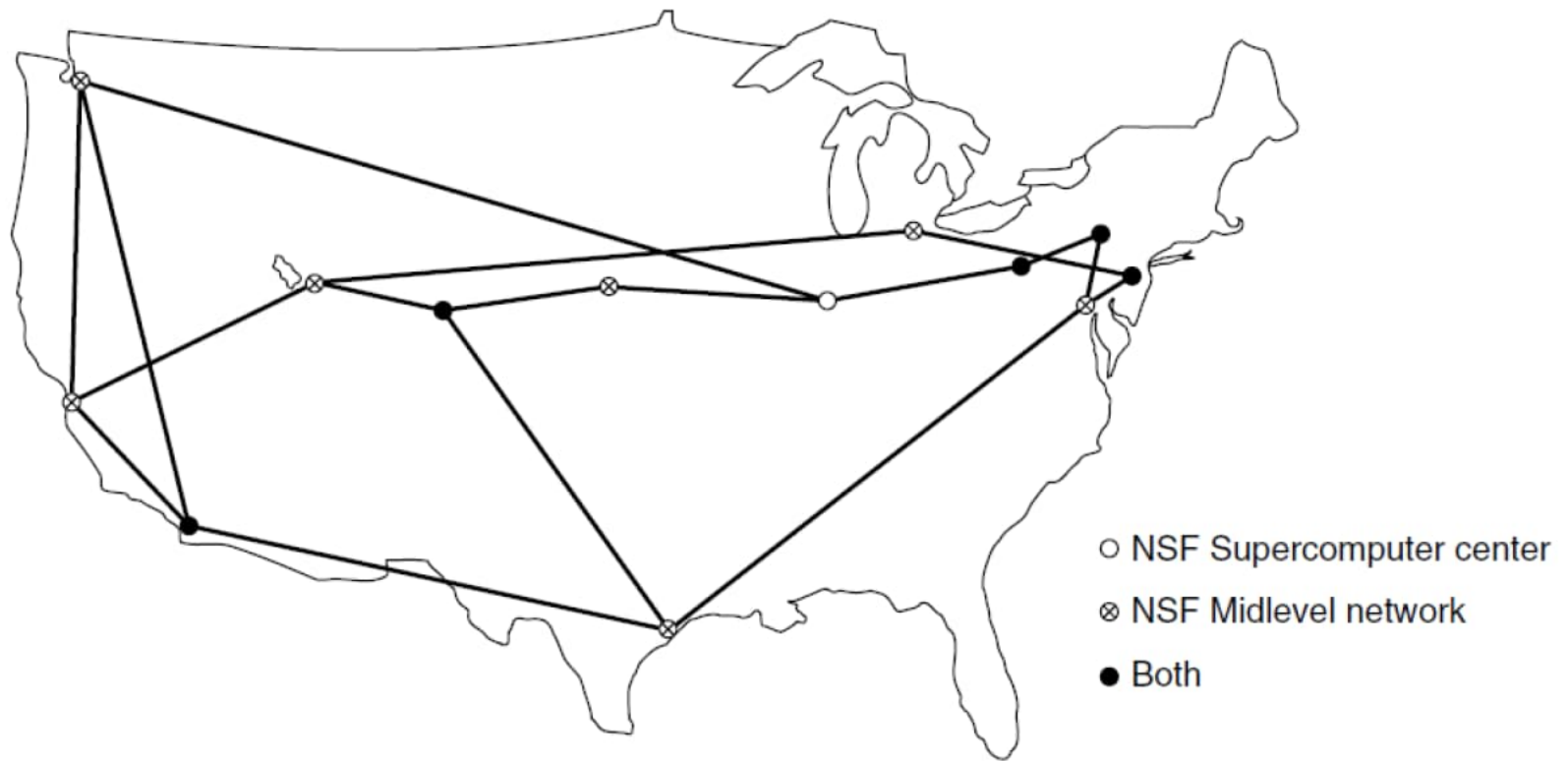


Figure 1-28. The NSFNET backbone in 1988.

Architecture of the Internet

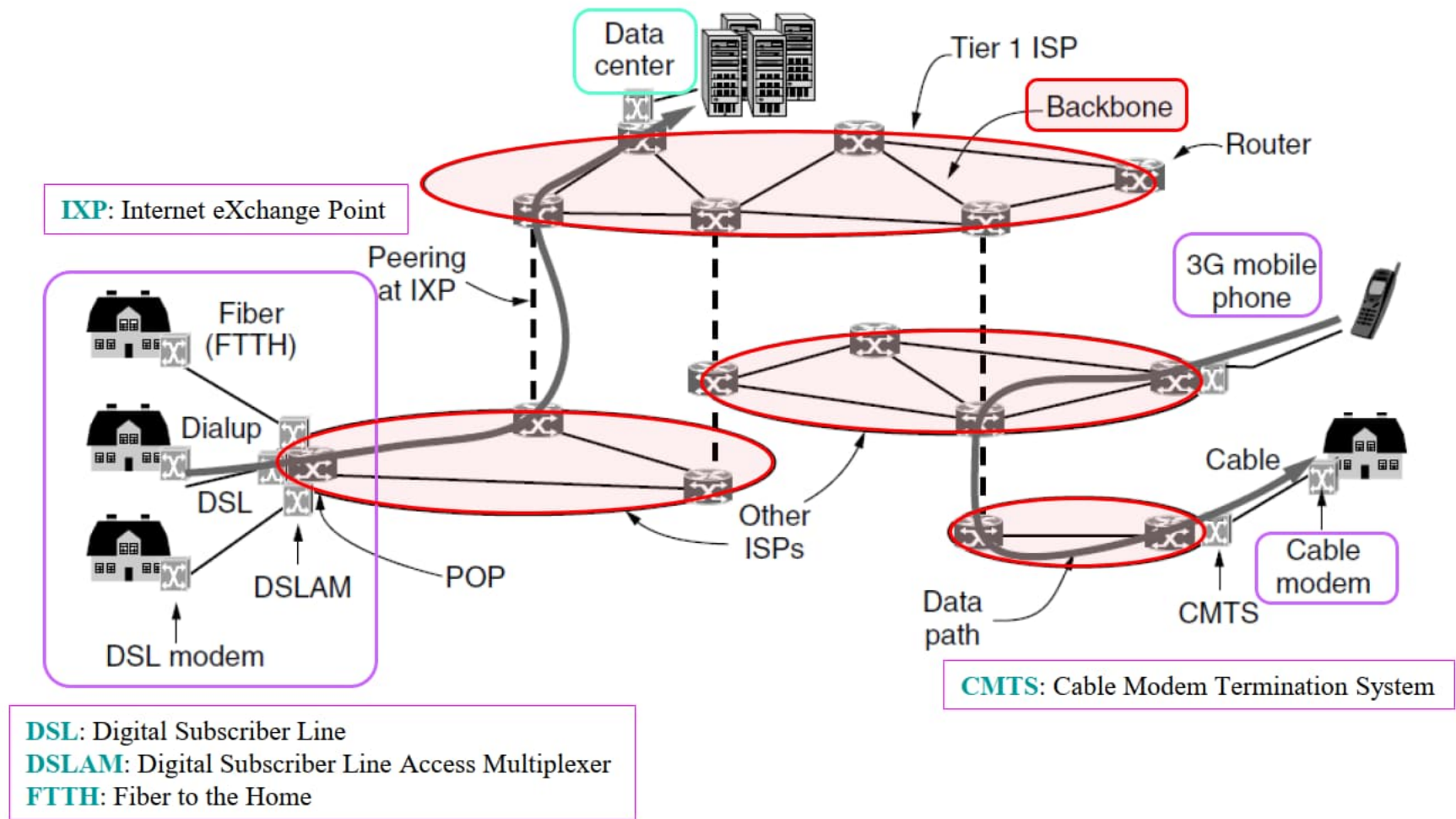


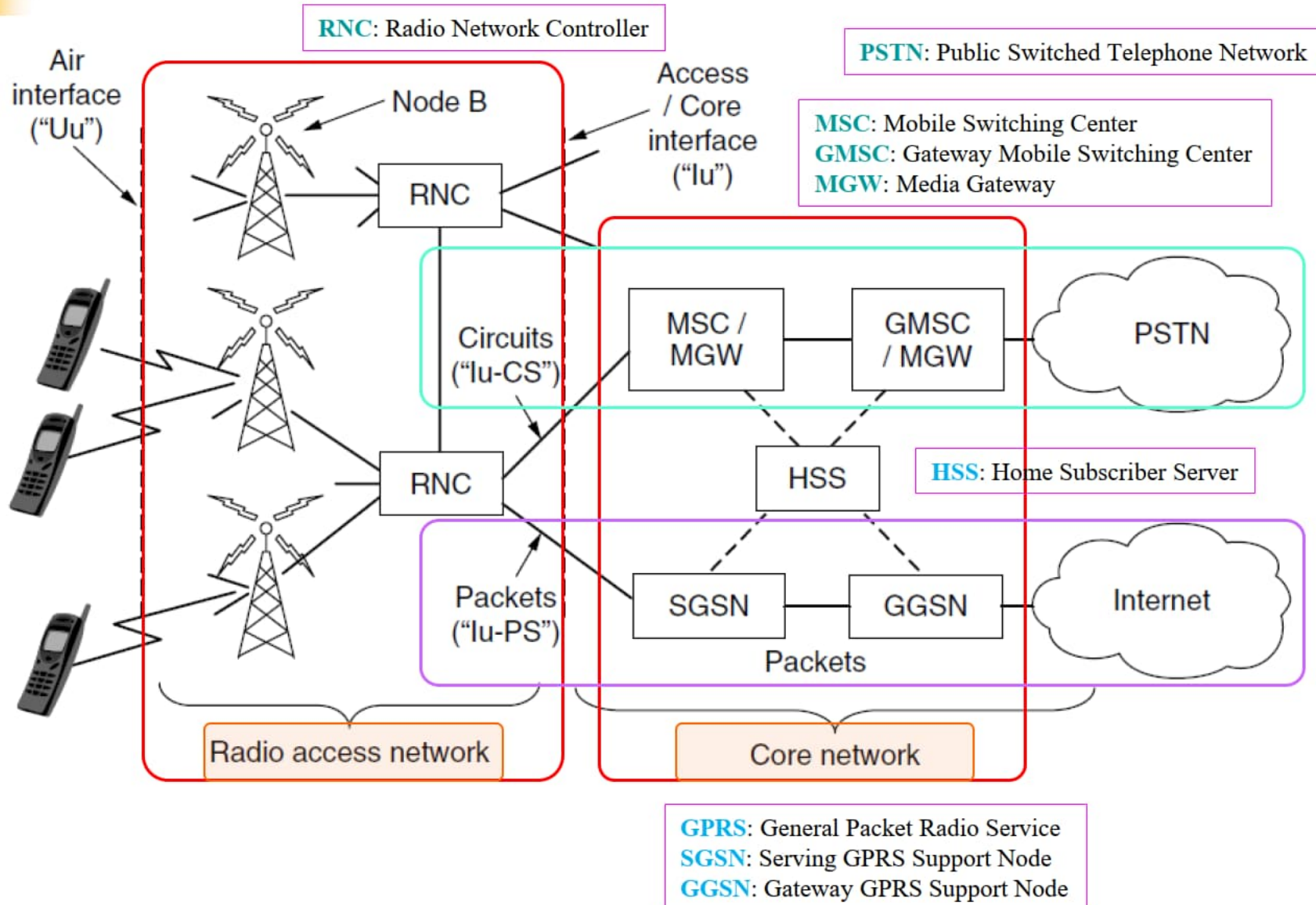
Figure 1-29. Overview of the Internet architecture.

1.5.2 Third-Generation Mobile Phone Networks



- First-generation
 - ✦ AMPS (Advanced Mobile Phone System)
- Second-generation
 - ✦ GSM (Global System for Mobile communications)
- The third generation
 - ✦ UMTS (Universal Mobile Telecommunications System)
 - ✦ WCDMA (Wideband Code Division Multiple Access)
- The fourth generation
 - ✦ LTE (Long Term Evolution)

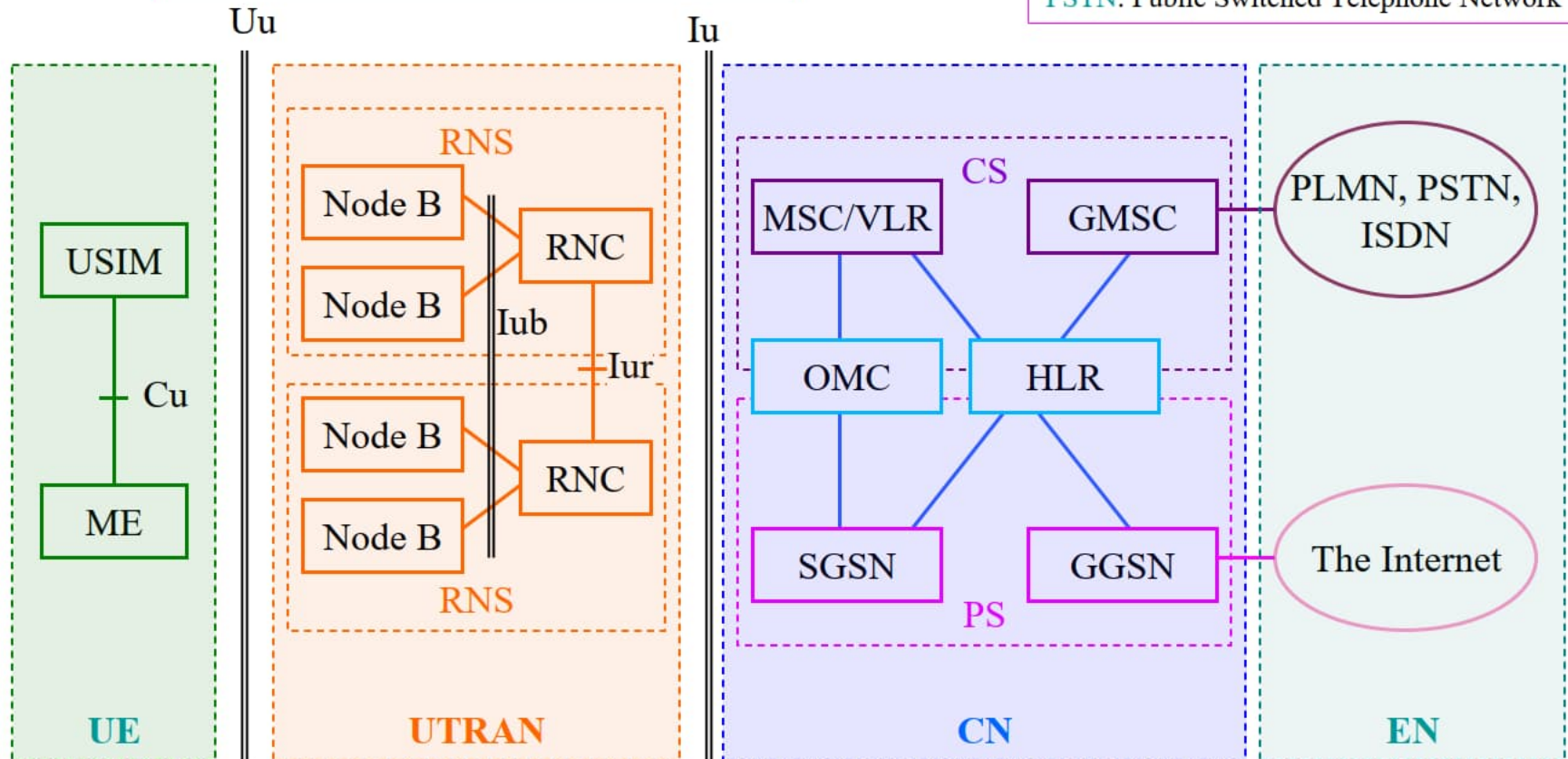
Architecture of the UMTS 3G mobile phone network



UMTS network units

RNC: Radio Network Controller
RNS: Radio Network Subsystem
UTRAN: UMTS Terrestrial Radio Access Network

PLMN: Public Land Mobile Network
PSTN: Public Switched Telephone Network

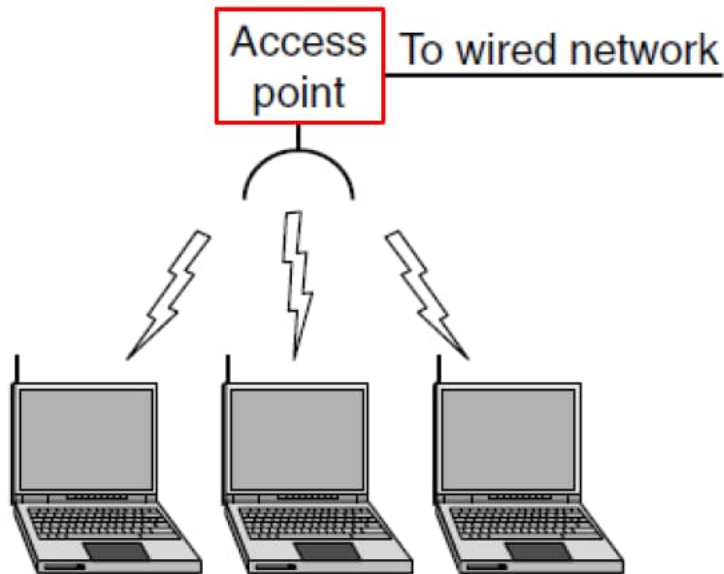


USIM: Universal Subscriber Identity Module
ME: Mobile Equipment
UE: User Equipment

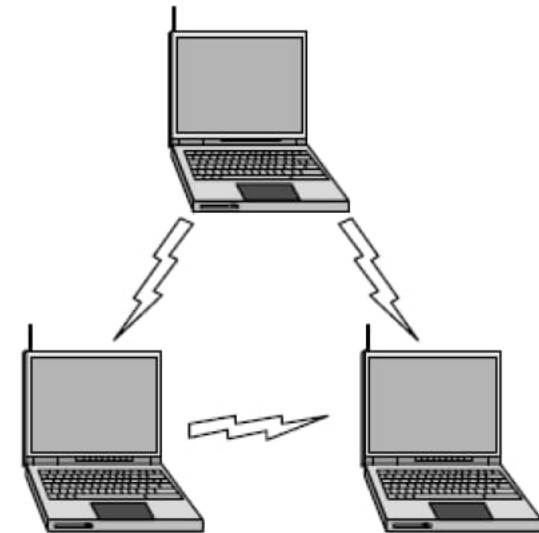
CS: Circuit Switching Domain
PS: Packet Switching Domain
CN: Core Network

EN: External Networks

1.5.3 Wireless LANs: 802.11



(a) Wireless network with an access point



(b) Ad hoc network

Figure 1-33

access point: 接入点
ad hoc: 自组织

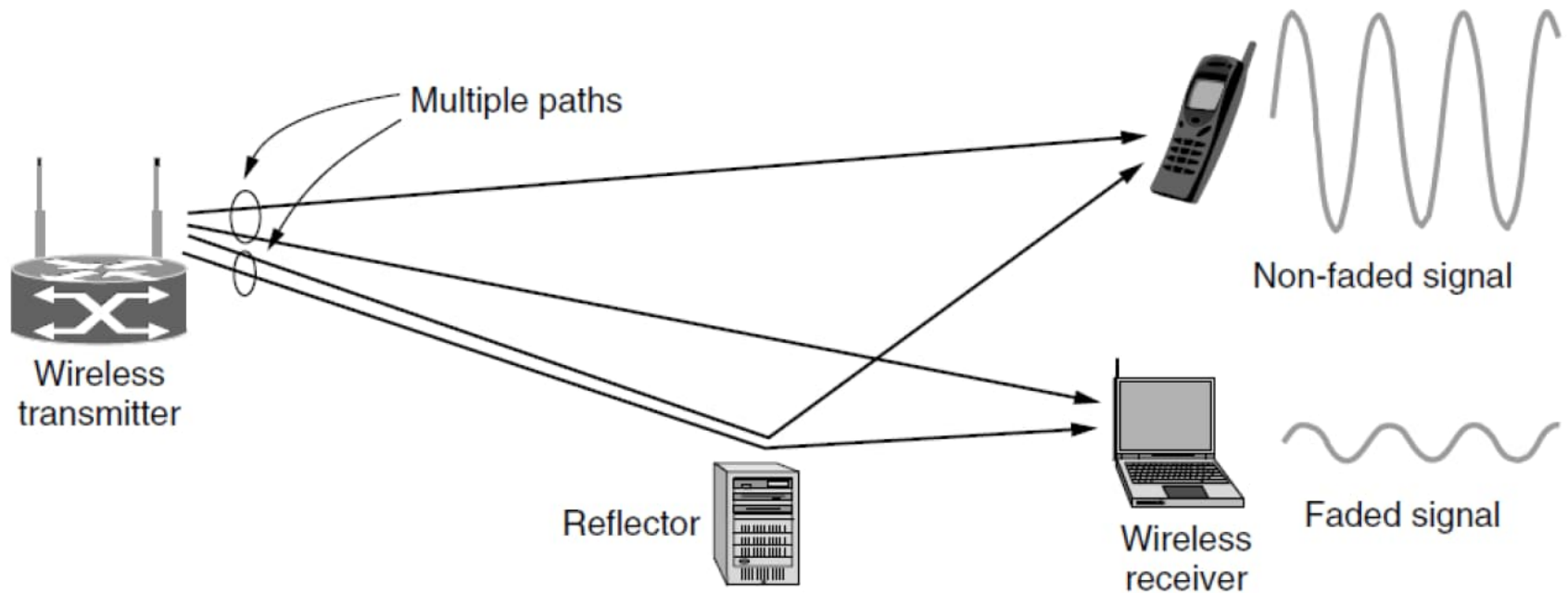


Figure 1-34. Multipath fading.

multipath fading: 多径衰落

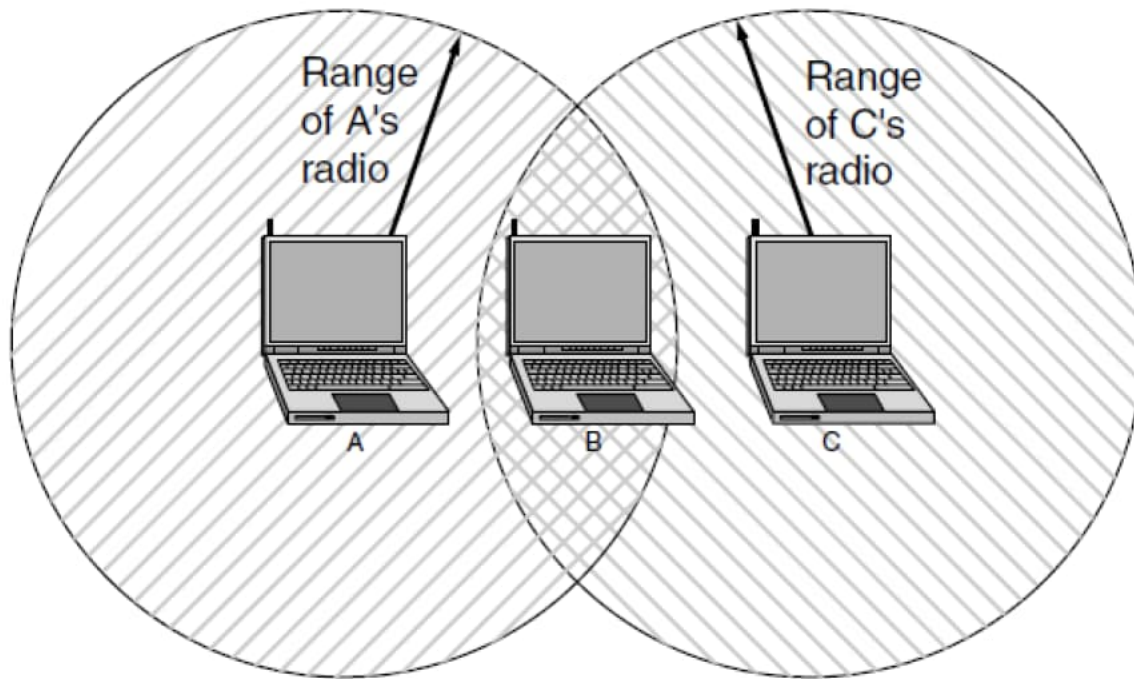


Figure 1-35. The range of a single radio may not cover the entire system.

1.5.4 RFID and Sensor Networks

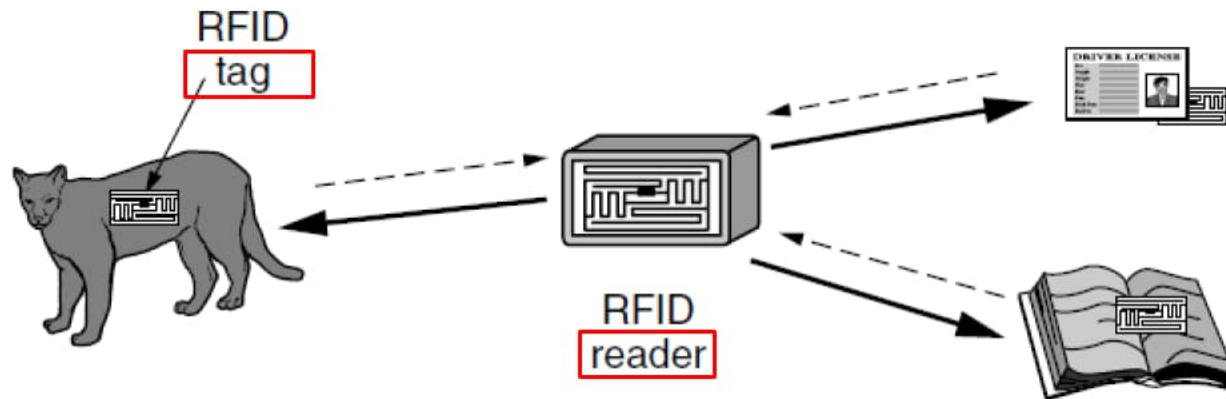


Figure 1-36. RFID used to network everyday objects.

Multihop topology

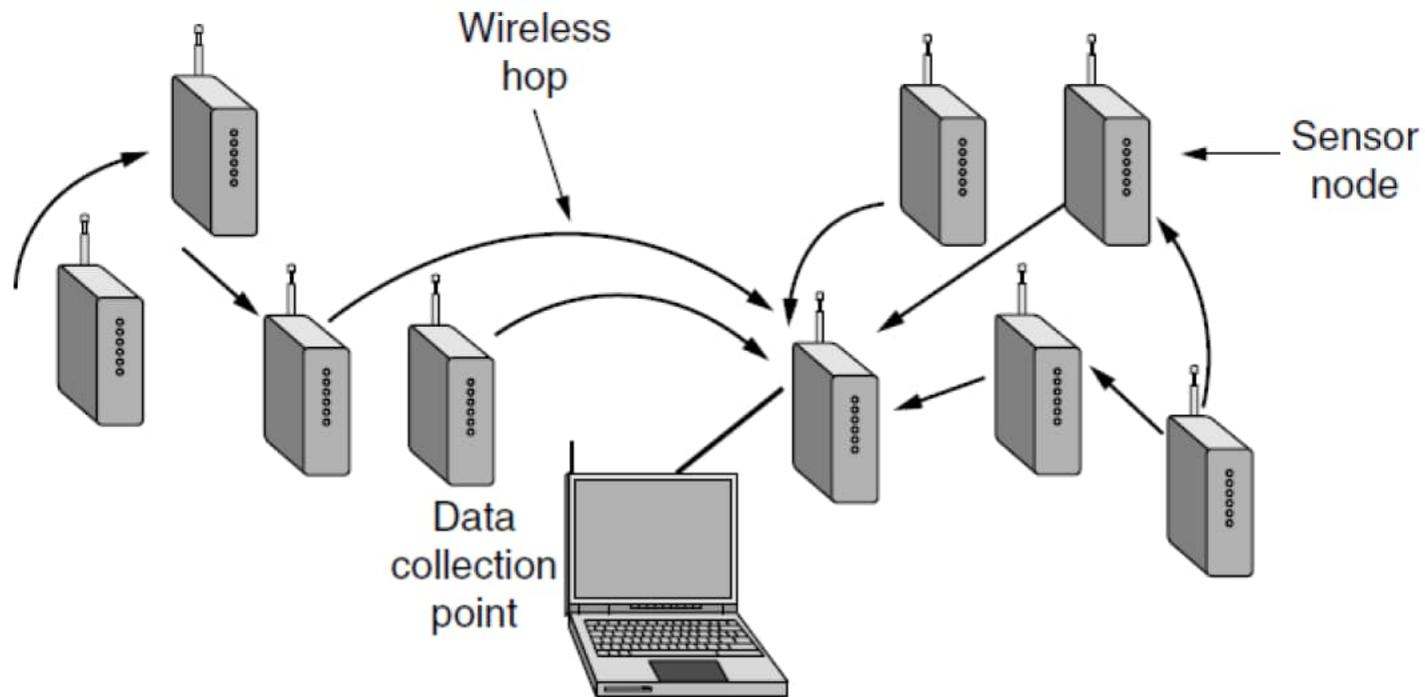


Figure 1-37. Multihop topology of a sensor network.

1.6 Network Standardization



1. Who's Who in the Telecommunications World
2. Who's Who in the International Standards World
3. Who's Who in the Internet Standards World

de facto and de jure



➤ Two Categories of Standards

✦ De facto, 事实标准

- Latin for “from the fact”
- just happened, without any formal plan.

de facto 英 [ˌdeɪ 'fæktəʊ] 美 [ˌdeɪ 'fæktou]
adj. <拉>实际的, 事实上的;
adv. 实际上, 事实上

✦ De jure, 法律标准

- Latin for “by law”
- adopted through the rules of some formal standardization body

de jure 英 [ˌdeɪ 'dʒʊəri] 美 [ˌdeɪ 'dʒʊri]
adj. 法律上; <拉>理上的, 权利上的

Who's Who



➤ Telecommunications World

- ✦ ITU: International Telecommunication Union, 国际电信联盟
 - ITU-T: telephone and data communication systems
 - H.264, X.509,

➤ International standards World

- ✦ ISO: International Standards Organization, 国际标准化组织
- ✦ NIST: National Institute of Standards and Technology, 国家标准和技术协会
- ✦ IEEE: Institute of Electrical and Electronics Engineers, 电气电子工程师协会

➤ Internet Standards World

- ✦ IAB: Internet Architecture Board, 网络互联体系结构委员会
 - RFCs: Request For Comments
- ✦ W3C: World Wide Web Consortium, 万维网联盟

Who's Who 名人录

The 802 working groups

* important
 ↓ hibernating
 † gave up

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs (WiFi)
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number; nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth, Zigbee)
802.16 *	Broadband wireless (WiMAX)
802.17	Resilient packet ring
802.18	Technical advisory group on radio regulatory issues
802.19	Technical advisory group on coexistence of all these standards
802.20	Mobile broadband wireless (similar to 802.16e)
802.21	Media independent handoff (for roaming over technologies)
802.22	Wireless regional area network

1.7 METRIC UNITS



Prefix	Exp.	Explicit	Prefix	Exp.	Explicit
milli	10^{-3}	0.001	Kilo	10^3	1 000
micro	10^{-6}	0.000 001	Mega	10^6	1 000 000
nano	10^{-9}	0.000 000 001	Giga	10^9	1 000 000 000
pico	10^{-12}	0.000 000 000 001	Tera	10^{12}	1 000 000 000 000
femto	10^{-15}	0.000 000 000 000 001	Peta	10^{15}	1 000 000 000 000 000
atto	10^{-18}	0.000 000 000 000 000 001	Exa	10^{18}	1 000 000 000 000 000 000
zepto	10^{-21}	0.000 000 000 000 000 000 001	Zetta	10^{21}	1 000 000 000 000 000 000 000
yocto	10^{-24}	0.000 000 000 000 000 000 000 001	Yotta	10^{24}	1 000 000 000 000 000 000 000 000
ronto	10^{-27}	0.000 000 000 000 000 000 000 000 001	Ronna	10^{27}	1 000 000 000 000 000 000 000 000 000
quecto	10^{-30}	0.000 000 000 000 000 000 000 000 000 001	Quetta	10^{30}	1 000 000 000 000 000 000 000 000 000 000

Figure 1-39. The principal metric prefixes.

Summary



- Computer networks have many uses, both
 - ✦ for companies and for individuals,
 - in the home and while on the move.
- Networks can be divided into
 - ✦ LANs, MANs, WANs, and internetworks.
- Network software is built around protocols.
 - ✦ Protocol stacks are typically based either on the OSI model or on the TCP/IP model.
 - Both have link, network, transport, and application layers,
 - but they differ on the other layers.
- Networks provide various services to their users.
 - ✦ connectionless best-efforts packet delivery
 - ✦ connection-oriented guaranteed delivery